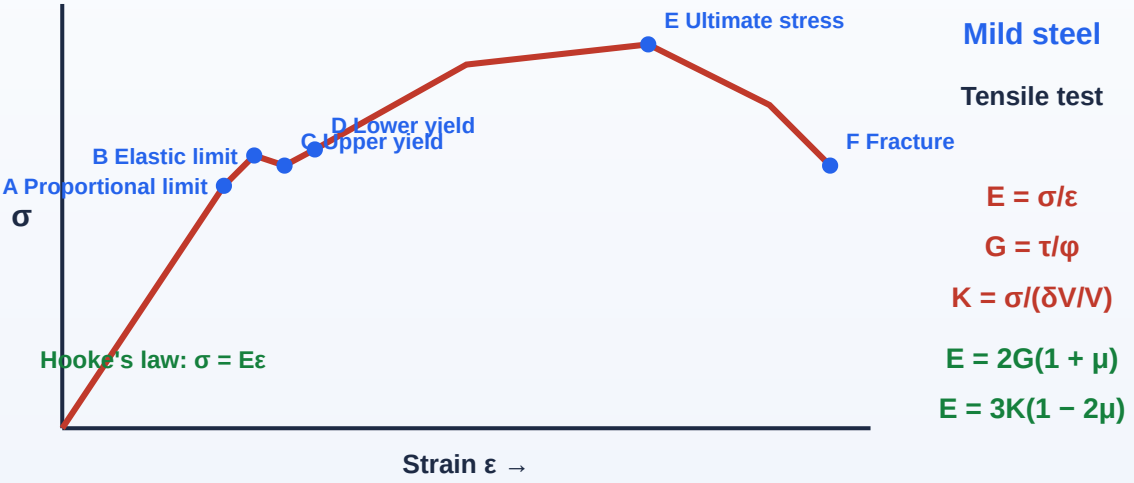
UPSSSC ALERT

UPSSSC JE

MECHANICAL STRENGTH OF MATERIALS

Complete English Edition

Unit 3 · 10 Marks · 2026



UPSSSC JE (Mechanical) 2026: Part-1 has **10 questions (10 marks)** from **Strength of Materials**. Syllabus: stress and strain, elastic limit and elastic constants, riveted joints, BM and SF diagrams, stress in composite bars, torsion of circular shafts and thin-walled pressure vessels.

Each chapter has key points, diagrams, formula table, detailed explanation, solved numericals, one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (R. K. Rajput, R. S. Khurmi, S. Ramamrutham).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Stress, Strain & Elastic Constants

Stress, Strain & Elastic Constants – Basic Concept

When an external force acts on a body, an internal resisting force develops. The resisting force per unit area is called **stress**, and the ratio of change in length to original length is called **strain**. Up to the **elastic limit**, the body regains its original shape when the load is removed.

Understand in Detail — Stress, Strain & Elastic Constants

1.1 Stress — the 'resistance' inside a material

Stretch a rubber band — you apply a force and the band resists it from inside. When an external force (load) acts on a body, its molecules are displaced and an **internal resisting force** develops. This internal resistance per unit area is called **stress (σ)**: **σ = P/A**, unit N/m² (Pa); in engineering usually **N/mm² = MPa**.

Types of stress: (1) **tensile** — the force pulls and lengthens the body (crane rope); (2) **compressive** — the force pushes and shortens it (building column); (3) **shear (τ)** — the force acts parallel to a surface and tries to slide one part over another (cutting paper with scissors, a rivet): τ = P/A, where A is the area along which sliding would occur. Normal stress (σ) acts perpendicular to the surface, shear stress (τ) parallel to it.

1.2 Strain and Hooke's law

Strain = change in dimension ÷ original dimension — it tells by what **fraction** the body has deformed; it has no unit. **Linear strain** e = δL/L; **shear strain** φ = angular distortion (radians); **volumetric strain** e_v = δV/V.

Hooke's law: within the elastic limit, stress is proportional to strain: **σ = E e**. The constant E = **Young's modulus of elasticity** — a measure of the material's 'stiffness'; the higher E, the less the stretching. Steel E ≈ 200 GPa, aluminium ≈ 70 GPa, rubber ≈ 0.01 GPa. For shear **τ = G φ**, G = **modulus of rigidity**; for volume **p = K e_v**, K = **bulk modulus**.

Derivation: Elongation of a bar δL = PL/AE

1. σ = P/A and e = δL/L
2. Hooke: P/A = E × δL/L
3. **δL = PL/(AE)** — more load or length gives more elongation; more area or E gives less
4. **Bar hanging under its own weight:** δL = wL²/(2E) = WL/(2AE) — half of that for the same weight hung at the end (w = specific weight)
5. **Tapering circular bar** (diameter d_1 to d_2): δL = 4PL/(π E d_1 d_2)

Example: A steel rod 2 m long, 20 mm diameter carries a tensile load of 50 kN. Stress, strain and elongation? E = 200 GPa.

Solution:

1. $A = (\pi/4)(20)^2 = 314.16\ mm^2$
2. $\sigma = 50000/314.16 =$ **159.2 N/mm²**
3. $e = \sigma/E = 159.2/200000 =$ **7.96×10^{-4}**
4. $\delta L = e \times L = 7.96 \times 10^{-4} \times 2000 =$ **1.59 mm**

1.3 Stress-strain curve — the 'biography' of mild steel

If a mild steel bar is slowly pulled to fracture in a **Universal Testing Machine (UTM)** and stress is plotted against strain, some special points appear (see figure):

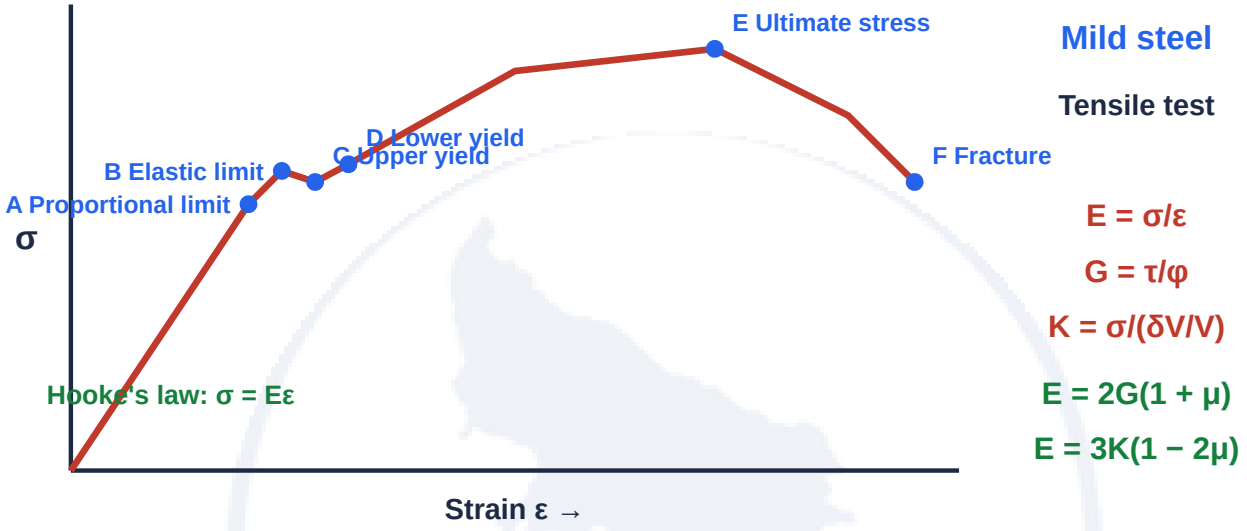


Fig 3.1: Stress–strain diagram of mild steel and elastic constants

(A) Proportional limit — up to here the graph is a straight line and Hooke's law holds. **(B) Elastic limit** — up to here the bar returns fully to its original length on removing the load. **(C, D) Upper and lower yield points** — suddenly, without increase of load, strain starts increasing; the material 'gives in'. **This yield strength is the most important in design. (E) Ultimate stress** — the maximum stress the material can bear; after this **necking** starts in the middle of the bar — the area falls rapidly. **(F) Breaking point** — the bar fractures. Because of necking the engineering curve appears to drop before fracture, while the **true stress** (on actual area) keeps rising.

Ductile vs brittle: ductile materials (mild steel, copper, aluminium) stretch a lot before fracture (percentage elongation > 5%) — they give warning; brittle materials (cast iron, glass, concrete) break suddenly without warning and have no yield point. For materials without a clear yield point (aluminium), a **proof stress** is taken by the **0.2% offset method**.

Factor of safety (FOS) = ultimate (or yield) stress ÷ working (allowable) stress. It is kept to allow for uncertainties like material defects, load estimation, shocks, etc. For ductile materials under steady load ≈ 2–3, more under shock loads.

1.4 Poisson's ratio and the relation between elastic constants

Stretch rubber lengthwise — it becomes thinner. Pulling any material in one direction makes it contract in the perpendicular directions. **Lateral strain ÷ longitudinal strain = Poisson's ratio (μ or ν)**. For metals 0.25–0.33, rubber ≈ 0.5, cork ≈ 0. Theoretical range: **0 ≤ μ ≤ 0.5** (at 0.5 the volume does not change at all — an incompressible material).

Derivation: Relation between the three moduli

1. **E = 2G(1 + μ)** — relation between Young's modulus and rigidity
2. **E = 3K(1 − 2μ)** — relation between Young's and bulk modulus
3. Eliminating μ: **E = 9KG/(3K + G)** — remember as '9KG over 3K + G'
4. **Volumetric strain of a bar:** $e_v = e(1 - 2\mu)$; **equal stress on all three faces of a cube:** $e_v = 3\sigma(1 - 2\mu)/E$
5. Example: steel E = 200 GPa, μ = 0.3 → G = 200/(2 × 1.3) = 76.9 GPa ; K = 200/(3 × 0.4) = 166.7 GPa

Strain energy: the work done in stretching is stored in the bar as energy (like a stretched spring) — $U = \frac{1}{2} P \delta L = \sigma^2 AL/(2E)$. **Resilience** = strain energy; **proof resilience** = maximum energy up to the elastic limit; **modulus of resilience** = per unit volume = $\sigma^2/2E$. A **suddenly applied** load produces **twice** the stress of a gradually applied one ($\sigma = 2P/A$); a load falling from height h (impact) produces even more: $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/(PL))}]$.

1. Key Points

- **Stress (σ):** $\sigma = P/A$; unit N/m² (Pa) or N/mm² (MPa). Types: tensile, compressive and shear stress.
- **Strain (ε):** $\varepsilon = \delta L/L$; dimensionless. Shear strain (φ) is angular distortion; volumetric strain = δV/V.
- **Hooke's law:** up to the proportional limit, stress is proportional to strain: $\sigma = E\varepsilon$, where E is Young's modulus.
- **Elastic constants:** Young's modulus $E = \sigma/\varepsilon$, modulus of rigidity $G = \tau/\varphi$, bulk modulus $K = \sigma/(\delta V/V)$, and Poisson's ratio μ = lateral strain/longitudinal strain.
- **Relations:** $E = 2G(1 + \mu)$, $E = 3K(1 - 2\mu)$ and $E = 9KG/(3K + G)$.
- **Poisson's ratio:** 0.25–0.33 for metals; theoretical maximum 0.5 (incompressible, e.g. rubber) and nearly 0 for cork.
- **Elongation of a bar:** $\delta L = PL/(AE)$. Elongation of a bar hanging under its own weight = $\rho g L^2/(2E) = WL/(2AE)$, half that of the same load at the end.
- **Factor of safety:** ultimate (or yield) stress ÷ working (permissible) stress. Yield stress is used for ductile and ultimate stress for brittle materials.
- **Shear stress (τ):** $\tau = P/A$ (A = area of shear plane). Complementary shear: a shear stress on a plane is always accompanied by an equal shear stress on the perpendicular plane.
- **Working stress:** the maximum safe stress = yield (or ultimate) stress ÷ factor of safety.

- **Material properties:** ductility, malleability, hardness, toughness, elasticity, plasticity and brittleness.
- **Impact load:** W falling through h gives $\sigma = (W/A)[1 + \sqrt{(1 + 2AEh/(WL))}]$; with h = 0, $\sigma = 2W/A$.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Stress / Strain	$\sigma = P/A$, $\varepsilon = \delta L/L$
Hooke's law	$\sigma = E\,\varepsilon$
Elongation	$\delta L = PL / AE$
E–G relation	$E = 2G(1 + \mu)$
E–K relation	$E = 3K(1 - 2\mu)$
E, G, K relation	$E = 9KG / (3K + G)$
Volumetric strain (bar)	$\delta V/V = \varepsilon(1 - 2\mu)$
Elongation under own weight	$\delta = \rho g L^2 / 2E$
Elongation of tapering bar	$\delta = 4PL / (\pi E d_1 d_2)$
Strain energy	$U = \sigma^2 V / 2E$

3. Detailed Explanation

- **Stress–strain diagram (mild steel):** proportional limit → elastic limit → upper and lower yield points → strain hardening → ultimate stress → necking → fracture.
- **Ductile and brittle materials:** ductile materials (mild steel, copper) show large plastic deformation and a clear yield point before fracture; brittle ones (cast iron, concrete) fail without warning.
- **Proof stress:** for materials without a distinct yield point (aluminium, high-carbon steel), yield is found by the 0.2% offset method.
- **Elastic constants:** an isotropic material has four elastic constants (E, G, K, μ), but only two are independent.
- **Strain energy and resilience:** energy stored up to elastic limit = ½ × P × δL. Maximum energy per unit volume is the modulus of resilience = $\sigma^2/2E$. A suddenly applied load produces twice the stress of a gradually applied one.
- **Percentage elongation and reduction in area:** measures of ductility in a tensile test; about 25–30% elongation for mild steel.
- **Fatigue and creep:** failure under repeated fluctuating loads even below the elastic limit is fatigue; slow increase of strain under constant load at high temperature is creep.
- **Stress concentration:** near holes, notches or sharp corners the local stress rises several times the average; it is reduced by providing fillets.

4. Comparison Table — Ductile vs Brittle

Basis	Ductile	Brittle
Deformation before fracture	Large	Very little
Yield point	Distinct	None
Design basis	Yield stress	Ultimate stress
Examples	Mild steel, copper	Cast iron, glass, concrete

5. Solved Numerical Problems

Example 1. A steel rod 2 m long and 20 mm diameter carries 31.4 kN tension. E = 200 GPa. Find stress and elongation.

Solution:

1. $A = \pi/4 \times 20^2 = 314.16\ mm^2$
2. $\sigma = 31400/314.16 =$ **$100\ N/mm^2$**
3. $\delta L = \sigma L/E = 100 \times 2000/200000 =$ **1 mm**

Example 2. If E = 200 GPa and μ = 0.3, find G and K.

Solution:

1. $G = E/2(1 + \mu) = 200/2.6 =$ **76.9 GPa**
2. $K = E/3(1 - 2\mu) = 200/1.2 =$ **166.7 GPa**

Example 3. Yield stress of mild steel is 250 MPa and factor of safety 2.5. Find the permissible stress.

Solution:

1. Permissible stress = 250/2.5 = **100 MPa**

Example 4. A 10 mm pin carries 15 kN in double shear. Find the shear stress.

Solution:

1. $\tau = P/(2 \times \pi d^2/4) = 15000/(2 \times 78.54)$
2. $\tau =$ **$95.5\ N/mm^2$**

Example 5. A 20 kN load is suddenly applied to a 2 m bar of area 400 mm^2. E = 200 GPa. Find stress and elongation.

Solution:

1. $\sigma = 2P/A = 2 \times 20000/400 =$ **$100\ N/mm^2$**
2. $\delta L = \sigma L/E = 100 \times 2000/200000 =$ **1 mm**

Example 6. A bar's length changes from 1000 to 1001.5 mm and diameter from 20 to 19.991 mm. Find Poisson's ratio.

Solution:

1. Longitudinal strain = 1.5/1000 = 0.0015; lateral strain = 0.009/20 = 0.00045
2. $\mu = 0.00045/0.0015 = \mathbf{0.3}$

6. Frequently Asked Facts

- $E = 2G(1 + \mu)$ and $E = 3K(1 - 2\mu)$ are the most asked relations.
- Maximum Poisson's ratio is 0.5; then K becomes infinite.
- Elongation under own weight is half that of the same load at the end.

7. Note (Confusing Facts)

- Elastic limit and proportional limit are different; Hooke's law holds only up to the proportional limit.
- Strain is dimensionless, but stress and E have the same unit (N/mm^2).
- A suddenly applied load doubles the stress; an impact load gives even more.

Remember: One line: E > K > G for common metals ($\mu \approx 0.3$). Assertion–Reason: (A) Rubber hardly changes volume. (R) Its Poisson's ratio is about 0.5. — Both true; R explains A.

30 Most Important One-Liner Facts — Stress, Strain & Elastic Constants

1. Unit of stress?

— N/m^2 (Pa).

2. Unit of strain?

— Dimensionless.

3. Hooke's law?

— $\sigma = E\varepsilon$ (up to proportional limit).

4. Unit of Young's modulus?

— N/mm^2.

5. E of steel approximately?

— 200 GPa.

6. Modulus of rigidity?

— $G = \tau/\varphi$.

7. Bulk modulus?

— $K = \sigma/(\delta V/V)$.

8. Poisson's ratio?

— Lateral strain ÷ longitudinal strain.

9. Maximum Poisson's ratio?

— 0.5.

10. Poisson's ratio of metals?

— 0.25–0.33.

11. Relation of E and G?

— $E = 2G(1 + \mu)$.

12. Relation of E and K?

— $E = 3K(1 - 2\mu)$.

13. Independent elastic constants?

— Two.

14. Elongation of a bar?

— PL/AE.

15. Elongation due to own weight?

— $\rho gL^2/2E$.

16. Local narrowing before fracture?

— Necking.

17. Design basis for ductile material?

— Yield stress.

18. Design basis for brittle material?

— Ultimate stress.

19. Stress due to sudden load?

— Twice the gradual load stress.

20. Modulus of resilience?

— $\sigma^2/2E$.

21. Shear stress?

— $\tau = P/A$ (shear plane).

22. Complementary shear?

— Equal shear on perpendicular plane.

23. Working stress?

— Yield ÷ factor of safety.

24. Failure under repeated loading?

— Fatigue.

25. Slow strain at high temperature?

— Creep.

26. Stress rise at sharp corners?

— Stress concentration.

27. Property to be drawn into wire?

— Ductility.

28. Property to be beaten into sheets?

— Malleability.

29. Resistance to fracture by absorbing energy?

— Toughness.

30. Percent elongation of mild steel?

— ≈ 25–30%.

Practice MCQs — Stress, Strain & Elastic Constants

Q1. Hooke's law holds up to:

(a) Proportional limit
(b) Yield point
(c) Ultimate stress
(d) Fracture

✔ **Answer: (a) Proportional limit**

Explanation: Up to proportional limit $\sigma \propto \varepsilon$.

Q2. Relation between E, G and μ:

(a) $E = 2G(1 + \mu)$
(b) $E = 3G(1 - 2\mu)$
(c) $E = G(1 + \mu)$
(d) $E = 2G(1 - \mu)$

✔ **Answer: (a) $E = 2G(1 + \mu)$**

Explanation: Standard relation.

Q3. Maximum theoretical value of Poisson's ratio:

(a) 0.25
(b) 0.33
(c) 0.5
(d) 1

✔ **Answer: (c) 0.5**

Explanation: At $\mu = 0.5$ volumetric strain is zero.

Q4. Independent elastic constants for an isotropic material:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

Explanation: Four constants, two independent.

Q5. Elongation of a bar hanging under its own weight:

(a) $\rho g L^2/E$ (b) $\rho g L^2/2E$

(c) $2\rho g L^2/E$ (d) $\rho g L/2E$

✔ **Answer: (b) $\rho g L^2/2E$**

Explanation: Half that of the same load at the end.

Q6. Factor of safety for ductile materials is based on:

(a) Ultimate stress (b) Yield stress

(c) Breaking stress (d) Proportional limit

✔ **Answer: (b) Yield stress**

Explanation: A ductile material is considered failed at yield.

Q7. Stress due to a suddenly applied load is ____ that of a gradually applied load:

(a) Half (b) Equal

(c) Twice (d) Four times

✔ **Answer: (c) Twice**

Explanation: $\sigma = 2P/A$.

Q8. E for mild steel is about:

(a) 70 GPa (b) 100 GPa

(c) 200 GPa (d) 400 GPa

✔ **Answer: (c) 200 GPa**

Explanation: $\approx 2 \times 10^5$ N/mm².

Q9. For materials without a distinct yield point, we use:

(a) Proof stress (b) Ultimate stress

(c) Breaking stress (d) Poisson's ratio

✔ **Answer: (a) Proof stress**

Explanation: 0.2% offset method.

Q10. In $E = 3K(1 - 2\mu)$, if $\mu = 0.5$, K is:

(a) Zero (b) E

(c) Infinite (d) E/3

✔ **Answer: (c) Infinite**

Explanation: The denominator becomes zero.

Q11. Failure under repeated fluctuating loads below the elastic limit:

(a) Creep (b) Fatigue

(c) Buckling (d) Impact

✔ **Answer: (b) Fatigue**

Explanation: Fatigue failure.

Q12. Property of a material to be drawn into wires:

(a) Malleability
(b) Ductility
(c) Hardness
(d) Brittleness

✔ **Answer: (b) Ductility**

Explanation: Ductility.

Q13. 10 mm pin in double shear with 15 kN — τ ≈

(a) 191 MPa
(b) 95.5 MPa
(c) 47.7 MPa
(d) 150 MPa

✔ **Answer: (b) 95.5 MPa**

Explanation: 15000/(2 × 78.54).

Q14. Slow increase of strain at high temperature under constant load:

(a) Fatigue
(b) Creep
(c) Elasticity
(d) Hardening

✔ **Answer: (b) Creep**

Explanation: Creep.

Q15. Complementary shear stress means:

(a) Equal shear on the perpendicular plane
(b) Double shear
(c) Tensile stress
(d) Zero stress

✔ **Answer: (a) Equal shear on the perpendicular plane**

Explanation: Needed for equilibrium.

SSC JE / UPSSSC JE Previous Year Questions — Stress, Strain & Elastic Constants

Q1. Relation between E, K and G:

SSC JE (Previous Year)

(a) E = 9KG/(3K + G)
(b) E = 3KG/(9K + G)
(c) E = 9KG/(K + 3G)
(d) E = KG/(3K + G)

✔ **Answer: (a) E = 9KG/(3K + G)**

Explanation: Standard relation.

Q2. In the stress–strain curve of mild steel, the first point reached is:

SSC JE (Previous Year)

(a) Elastic limit
(b) Proportional limit
(c) Yield point
(d) Ultimate point

✔ **Answer: (b) Proportional limit**

Explanation: Order: proportional → elastic → yield.

Q3. The unit of strain is: SSC JE (Previous Year)

(a) N/m^2 (b) m

(c) No unit (d) N

✔ **Answer: (c) No unit**

Explanation: Ratio of lengths.

Q4. If Poisson's ratio is 0.25, E/G equals: SSC JE (Previous Year)

(a) 2.0 (b) 2.5

(c) 3.0 (d) 1.5

✔ **Answer: (b) 2.5**

Explanation: E/G = 2(1 + 0.25) = 2.5.

Q5. A material that fractures without plastic deformation is: SSC JE (Previous Year)

(a) Ductile (b) Brittle

(c) Malleable (d) Tough

✔ **Answer: (b) Brittle**

Explanation: Brittle.

Q6. Necking in the stress–strain curve begins at: SSC JE (Previous Year)

(a) Elastic limit (b) Yield point

(c) Ultimate stress point (d) Fracture point

✔ **Answer: (c) Ultimate stress point**

Explanation: Necking starts after ultimate stress.

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Chapter 2: Riveted Joints

Riveted Joints – Basic Concept

A **rivet** is a permanent fastener used to join plates in boilers, tanks, bridges and ships. It has a head, shank and tail; it is inserted hot and the second head is formed by hammering.

Understand in Detail — Riveted Joints

2.1 Riveted joints — how two plates are joined

In old bridges, boilers, ships and storage tanks plates were joined by **rivets** (welding is more common today). A **rivet** is a short cylindrical rod with a **head** formed at one end. It is heated, inserted in the hole through the plates and the other end is hammered to form a second head. On cooling the rivet contracts and grips the plates tightly. It is a **permanent joint** — the rivet must be cut to open it.

Types of joints: (1) **lap joint** — one plate overlaps the other; (2) **butt joint** — the plates meet edge to edge with a **cover plate (strap)** on one or both sides; with a single cover the rivets are in **single shear**, with double covers in **double shear**. By rows of rivets — single, double, triple riveted; arrangement — **chain** or **zig-zag**.

Terminology: pitch (p) — distance between centres of adjacent rivets in a row; **margin (m)** — from rivet centre to plate edge ≈ 1.5d; **back pitch** — distance between two rows; **diagonal pitch** — oblique distance in zig-zag. **Unwin's formula** — rivet diameter **d = 6√t** (in mm, t = plate thickness, when t > 8 mm).

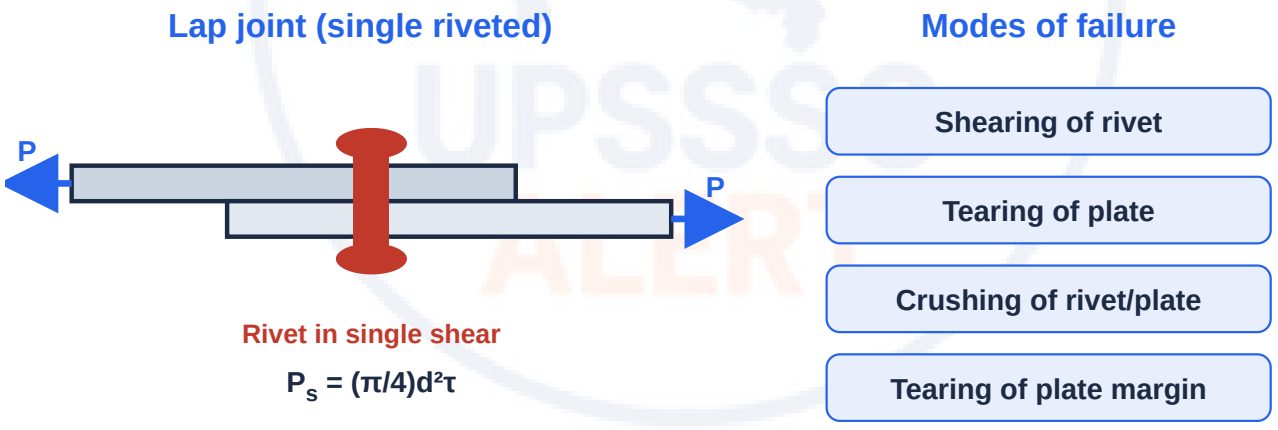


Fig 3.2: Riveted lap joint and modes of failure

2.2 How a joint fails — and its strength

Consider one pitch length (the strip of plate belonging to one rivet). The joint can fail in four ways:

Derivation: Modes of failure and strength (per pitch)

1. **(1) Tearing of plate** — along the row of holes, where the plate is weakest: $\mathbf{P_t = (p - d)\, t\, \sigma_t}$
2. **(2) Shearing of rivet** — the rivet is cut into two: single shear $P_s = n\,(\pi/4)\,d^2\,\tau$; double shear $P_s = n \times 2 \times (\pi/4)\,d^2\,\tau$ (Indian standard uses 1.75 times in practice)
3. **(3) Crushing (bearing)** — the rivet or plate hole gets crushed oval: $\mathbf{P_c = n\, d\, t\, \sigma_c}$ (projected area $d \times t$)
4. **(4) Tearing of margin** — the rivet tears out through the plate edge; prevented by keeping $m \geq 1.5d$
5. **Strength of joint** = the **least** of P_t, P_s, P_c (the weakest link decides the chain's strength)
6. **Efficiency of joint η = strength of joint ÷ strength of solid plate without holes ($p\, t\, \sigma_t$)** ; based on tearing alone: $\eta = (p - d)/p$

Balanced joint — one in which tearing, shearing and crushing strengths are nearly equal; such a design is the most economical. Typical efficiencies — single-riveted lap ≈ 50–60%, double-riveted ≈ 70–75%, triple-riveted double-cover butt ≈ 85–90%. **Caulking and fullering** — in pressure vessels like boilers the plate edge is hammered with a blunt tool to make the joint leak-proof.

Example: Single-riveted lap joint of 12 mm plates; rivet d = 20 mm, pitch 60 mm; $\sigma_t = 120$, $\tau = 90$, $\sigma_c = 180$ N/mm². Strength and efficiency?

Solution:

1. Tearing: $(60 - 20) \times 12 \times 120 =$ **57600 N**
2. Shearing (single): $(\pi/4)(20)^2 \times 90 = 314.16 \times 90 =$ **28274 N**
3. Crushing: $20 \times 12 \times 180 =$ **43200 N**
4. Strength = least = 28274 N (shearing) ; solid plate = $60 \times 12 \times 120 = 86400$ N
5. $\eta = 28274/86400 =$ **32.7%** — very low! The rivet is weak; improve with double shear or bigger rivets

1. Key Points

- **Types of joints:** lap joint (plates overlap) and butt joint (plates end to end with one or two cover plates).
- **Rows of rivets:** single, double or triple riveted; in multiple rows, chain or zig-zag arrangement.
- **Pitch (p):** distance between centres of adjacent rivets in a row. **Margin (m):** distance from rivet centre to plate edge ≈ 1.5d.
- **Unwin's formula:** rivet hole diameter $d = 6\sqrt{t}$ (mm), where t is plate thickness in mm.
- **Modes of failure:** (i) tearing of plate: $P_t = (p - d)t\sigma_t$; (ii) shearing of rivet: $P_s = n(\pi/4)d^2\tau$; (iii) crushing: $P_c = n \cdot d \cdot t \cdot \sigma_c$.
- **Shear:** in a lap joint and single-cover butt joint the rivet is in single shear; in a double-cover butt joint it is in double shear (strength taken as 1.875 times, IBR).

- **Efficiency of joint:** η = least of P_t, P_s, P_c ÷ strength of solid plate ($p \cdot t \cdot \sigma_t$). A riveted joint is never 100% efficient.
- **Caulking and fullering:** plate edges are pressed with a caulking tool to make the joint leak-proof; fullering gives a better seal.
- **Pitch values:** minimum pitch ≈ 2.5d–3d (room for hammering); maximum pitch limited by leak-tightness and buckling.
- **Back pitch:** distance between two consecutive rows; **diagonal pitch:** diagonal distance between rivets of adjacent rows in zig-zag riveting.
- **Welded vs riveted:** welding is lighter, leak-proof and more efficient (≈ 100%); riveting suits dissimilar metals and vibrating structures.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Unwin's formula	$d = 6\sqrt{t}$ (mm)
Tearing of plate	$P_t = (p - d)\, t\, \sigma_t$
Shearing of rivet (single)	$P_s = n\, (\pi/4)\, d^2\, \tau$
Shearing of rivet (double)	$P_s = 2n\, (\pi/4)\, d^2\, \tau$
Crushing	$P_c = n\, d\, t\, \sigma_c$
Strength of solid plate	$P = p\, t\, \sigma_t$
Efficiency	$\eta = \min(P_t, P_s, P_c) / (p\, t\, \sigma_t)$
Tearing efficiency	$\eta_t = (p - d)/p$

3. Detailed Explanation

- **Best design:** a joint is most balanced when its tearing, shearing and crushing strengths are nearly equal.
- **Joints in boilers:** lap joints are used for circumferential seams and butt joints for longitudinal seams, because the hoop stress is double.
- **Types of rivet heads:** snap (button) head is the most common; countersunk head where a flush surface is needed; pan and conical heads are also used.
- **Rivet material:** ductile — mild steel, wrought iron, copper, aluminium — so that the head can be formed by hammering.
- **Double-riveted joint calculation:** with n = 2 rivets per pitch, $P_s = 2(\pi/4)d^2\tau$ and $P_c = 2dt\sigma_c$; tearing only along the outer row $(p - d)t\sigma_t$.
- **Ways to increase efficiency:** more rows, double cover plates, diamond riveting (fewer rivets in the outer row) — these raise the tearing strength.

4. Comparison Table — Lap Joint vs Butt Joint

Basis	Lap joint	Butt joint
Plate position	Overlapping	End to end + cover plate
Rivet shear	Single	Double (with two covers)
Strength/efficiency	Lower	Higher
Use in boilers	Circumferential seam	Longitudinal seam

5. Solved Numerical Problems

Example 1. Find the rivet diameter for a 16 mm plate by Unwin's formula.

Solution:

1. $d = 6\sqrt{16} = 6 \times 4 =$ **24 mm**

Example 2. Single riveted lap joint: p = 60 mm, d = 20 mm, t = 10 mm, $\sigma_t = 100$, $\tau = 80$, $\sigma_c = 160$ N/mm² (one rivet per pitch). Find the efficiency.

Solution:

1. $P_t = (60 - 20) \times 10 \times 100 = 40{,}000$ N
2. $P_s = (\pi/4) \times 400 \times 80 = 25{,}133$ N
3. $P_c = 20 \times 10 \times 160 = 32{,}000$ N
4. Solid plate $= 60 \times 10 \times 100 = 60{,}000$ N
5. $\eta = 25{,}133/60{,}000 =$ **41.9%**

Example 3. Double-riveted lap joint: p = 80, d = 20, t = 10 mm, $\sigma_t = 100$, $\tau = 75$, $\sigma_c = 150$ N/mm². Find the efficiency.

Solution:

1. $P_t = (80 - 20) \times 10 \times 100 = 60{,}000$ N
2. $P_s = 2 \times (\pi/4) \times 400 \times 75 = 47{,}124$ N
3. $P_c = 2 \times 20 \times 10 \times 150 = 60{,}000$ N
4. Solid $= 80 \times 10 \times 100 = 80{,}000$ N
5. $\eta = 47{,}124/80{,}000 =$ **58.9%**

Example 4. Find Unwin's diameter and minimum pitch (3d) for a 25 mm plate.

Solution:

1. $d = 6\sqrt{25} =$ **30 mm**
2. Minimum pitch $= 3 \times 30 =$ **90 mm**

6. Frequently Asked Facts

- Unwin's formula $d = 6\sqrt{t}$ is frequently asked.
- Efficiency = least strength ÷ strength of solid plate.

- Butt joint for longitudinal seam of a boiler.

7. Note (Confusing Facts)

- Caulking makes a joint leak-proof, not stronger.
- Tearing uses (p − d) because the hole reduces the plate area.
- Crushing area is d × t (projected area), not $\pi d^2/4$.

Remember: One line: $d = 6\sqrt{t}$; $\eta = (p - d)/p$ for tearing. Assertion–Reason: (A) The longitudinal seam of a boiler is made stronger. (R) Hoop stress is twice the longitudinal stress. — Both true; R explains A.

30 Most Important One-Liner Facts — Riveted Joints

1. Rivet is which fastener?

— Permanent.

2. Unwin's formula?

— $d = 6\sqrt{t}$.

3. Pitch?

— Distance between adjacent rivet centres.

4. Margin?

— ≈ 1.5d.

5. Overlapping plates joint?

— Lap joint.

6. Plates end to end?

— Butt joint.

7. Rivet shear in lap joint?

— Single.

8. Double-cover butt joint?

— Double shear.

9. Tearing formula?

— $(p - d)t\sigma_t$.

10. Crushing formula?

— $n\ d\ t\ \sigma_c$.

11. Strength of solid plate?

— $p\ t\ \sigma_t$.

12. Efficiency of joint?

— Least strength ÷ solid plate.

13. Process to make leak-proof?

— Caulking / fullering.

14. Most common rivet head?

— Snap (button) head.

15. Head for flush surface?

— Countersunk.

16. Property of rivet material?

— Ductility.

17. Boiler longitudinal seam?

— Butt joint.

18. Boiler circumferential seam?

— Lap joint.

19. Can efficiency be 100%?

— No.

20. Efficiency in tearing?

— $(p - d)/p$.

21. Minimum pitch?

— ≈ 2.5d–3d.

22. Distance between rows?

— Back pitch.

23. Diagonal distance in zig-zag?

— Diagonal pitch.

24. Efficiency of welded joint?

— ≈ 100%.

25. Joining dissimilar metals?

— Riveting.

26. Benefit of diamond riveting?

— Higher efficiency.

27. Rivet inserted in which state?

— Red hot.

28. Parts of rivet?

— Head, shank, tail.

29. Rivets per pitch in double riveting?

— 2.

30. Double shear factor (IBR)?

— 1.875.

Practice MCQs — Riveted Joints

Q1. Rivet diameter by Unwin's formula:

(a) $d = 6\sqrt{t}$

(b) $d = 1.5t$

(c) $d = \sqrt{(6t)}$

(d) $d = 6t$

✔ **Answer: (a) $d = 6\sqrt{t}$**

Explanation: $d = 6\sqrt{t}$ (mm).

Q2. In a lap joint the rivet is in:

(a) Single shear

(b) Double shear

(c) Torsion

(d) Bending

✔ **Answer: (a) Single shear**

Explanation: One shear plane between two plates.

Q3. Tearing strength of plate:

(a) $(p - d)t\sigma_t$

(b) $p\ t\ \sigma_t$

(c) $d\ t\ \sigma_c$

(d) $(\pi/4)d^2\tau$

✔ **Answer: (a) $(p - d)t\sigma_t$**

Explanation: Area between holes.

Q4. To make a riveted joint leak-proof:

(a) Caulking

(b) Drilling

(c) Reaming

(d) Punching

✔ **Answer: (a) Caulking**

Explanation: Edges pressed to seal.

Q5. Rivet diameter for 16 mm plate (Unwin):

(a) 16 mm

(b) 20 mm

(c) 24 mm

(d) 32 mm

✔ **Answer: (c) 24 mm**

Explanation: $6\sqrt{16} = 24$.

Q6. Crushing area of a rivet:

(a) $\pi d^2/4$

(b) $d \times t$

(c) $p \times t$

(d) $(p - d) \times t$

✔ **Answer: (b) $d \times t$**

Explanation: Projected area.

Q7. In a double-cover butt joint the rivet is in:

(a) Single shear
(b) Double shear
(c) Crushing only
(d) Tension

✔ **Answer: (b) Double shear**

Explanation: Two shear planes.

Q8. Efficiency of a joint is based on:

(a) Maximum strength
(b) Least strength
(c) Average strength
(d) Crushing strength only

✔ **Answer: (b) Least strength**

Explanation: A joint fails in its weakest mode.

Q9. p = 50 mm, d = 20 mm — tearing efficiency:

(a) 40%
(b) 60%
(c) 70%
(d) 80%

✔ **Answer: (b) 60%**

Explanation: (50 − 20)/50 = 0.6.

Q10. Joint used for the longitudinal seam of a boiler:

(a) Lap
(b) Butt
(c) Any
(d) None

✔ **Answer: (b) Butt**

Explanation: Stronger butt joint for higher hoop stress.

Q11. Distance between two rows of rivets is called:

(a) Pitch
(b) Back pitch
(c) Margin
(d) Diagonal pitch

✔ **Answer: (b) Back pitch**

Explanation: Back pitch.

Q12. Minimum pitch of rivets is generally:

(a) d
(b) 1.5d
(c) 2.5d–3d
(d) 6d

✔ **Answer: (c) 2.5d–3d**

Explanation: Room for hammering.

Q13. Compared to a welded joint, a riveted joint is:

(a) More efficient
(b) Less efficient
(c) Lighter
(d) More leak-proof

✔ **Answer: (b) Less efficient**

Explanation: Holes weaken it.

Q14. Rivet material should be:

(a) Brittle
(b) Ductile
(c) Hard
(d) Cast

✔ **Answer: (b) Ductile**

Explanation: To form the head.

Q15. Rivets per pitch length in a double-riveted lap joint:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

Explanation: Two rows.

SSC JE / UPSSSC JE Previous Year Questions — Riveted Joints

Q1. Efficiency of a riveted joint is:

SSC JE (Previous Year)

(a) More than 100%
(b) Always less than 100%
(c) 100%
(d) Zero

✔ **Answer: (b) Always less than 100%**

Explanation: Holes weaken the plate.

Q2. Margin of a rivet is generally:

SSC JE (Previous Year)

(a) 0.5d
(b) d
(c) 1.5d
(d) 3d

✔ **Answer: (c) 1.5d**

Explanation: m ≈ 1.5d.

Q3. Purpose of caulking:

SSC JE (Previous Year)

(a) Strengthen the joint
(b) Make joint leak-proof
(c) Remove rivets
(d) Prevent rust

✔ **Answer: (b) Make joint leak-proof**

Explanation: Leak-proof.

Q4. Rivets for boiler plates are generally:

SSC JE (Previous Year)

(a) Snap head
(b) Pan head
(c) Countersunk
(d) Flat head

✔ **Answer: (a) Snap head**

Explanation: Snap/button head is common.

Q5. Rivet hole diameter compared to rivet diameter:

SSC JE (Previous Year)

(a) Slightly larger
(b) Slightly smaller
(c) Equal
(d) Double

✔ **Answer: (a) Slightly larger**

Explanation: ≈ 1.5 mm larger.

UPSSSC ALERT

Chapter 3: Shear Force & Bending Moment Diagrams

Shear Force & Bending Moment Diagrams – Basic Concept

A **beam** is a structural member loaded perpendicular to its axis. At a section, the algebraic sum of vertical forces on one side is the **shear force**, and the sum of their moments is the **bending moment**. Their plots are the SFD and BMD.

Understand in Detail — Shear Force & Bending Moment Diagrams

3.1 Beams, supports and loads — understand the picture first

A **beam** is a horizontal structural member that carries loads perpendicular to its length and transfers them to the supports by **bending** — like a roof joist, a bridge girder, a cupboard shelf.

Supports: simple/roller support — only vertical reaction (1 unknown); **pin (hinge)** — vertical + horizontal (2 unknowns); **fixed** — vertical + horizontal + moment (3 unknowns). **Types of beams: cantilever** — one end fixed in a wall, other free (a balcony); **simply supported** — simple supports at both ends; **overhanging** — part projecting beyond a support; **fixed and continuous** — indeterminate.

Loads: point load — concentrated at one point (kN); **uniformly distributed load (UDL)** — spread equally over the length (kN/m), taken in calculations as its total (w × length) acting at its mid-point; **uniformly varying load (UVL)** — triangular, total = ½ w L, centroid at L/3 from the heavy end.

3.2 Shear force and bending moment — what they are and how to find them

Imagine cutting the beam at a section X. To keep the left part in equilibrium two things are needed at the cut face: a vertical force — the **shear force (SF)**, and a turning effect — the **bending moment (BM)**. In simple words: **SF_X = algebraic sum of all vertical forces on one side of X; BM_X = algebraic sum of moments of all forces on one side of X about X.**

Sign convention: SF — upward force on the left (or downward on the right) is positive. BM — one causing **sagging** (bending down in the middle, a smile ☺ shape) is positive; **hogging** (bulging up) is negative. A cantilever has hogging BM, a simply supported beam sagging.

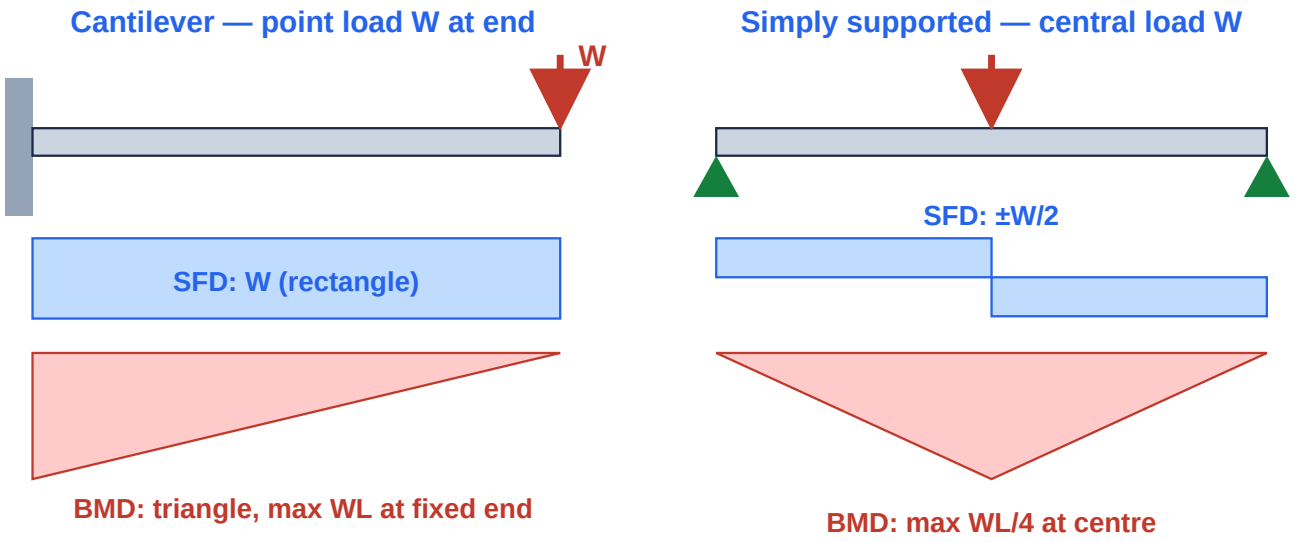


Fig 3.3: SF and BM diagrams of a cantilever and a simply supported beam

Derivation: Relation between load, SF and BM — the key to drawing diagrams

1. **dF/dx = −w** : slope of the SF diagram = load intensity. So — no load → SF horizontal line; UDL → SF inclined straight line; UVL → SF parabola (2nd degree)
2. **dM/dx = F** : slope of the BM diagram = SF. So — no load → BM inclined line; UDL → BM parabola; UVL → BM cubic curve (3rd degree)
3. **Most useful result: where SF = 0 (or changes sign), there BM is maximum** (dM/dx = 0)
4. Under a point load SF jumps suddenly; an external couple applied at a point makes BM jump suddenly
5. **Point of contraflexure** — where BM becomes zero and changes sign (only in overhanging/fixed beams)

Standard results — remember: cantilever, W at free end: M_{max} = WL (at fixed end); cantilever, full UDL w: $M_{max} = wL^2/2$; simply supported, W at centre: M_{max} = WL/4 (at centre), reactions W/2; simply supported, full UDL: $M_{max} = wL^2/8$ (at centre), SF_{max} = wL/2 (at supports); simply supported, W at distances a and b: M_{max} = Wab/L.

Example: A 6 m simply supported beam carries a 30 kN point load at 2 m from the left support. Reactions, SF and maximum BM?

Solution:

1. Moments about B: $R_A \times 6 = 30 \times 4 \rightarrow R_A =$ **20 kN** ; $R_B = 30 - 20 =$ **10 kN**
2. SF: +20 kN from A to the load, after the load 20 − 30 = −10 kN (up to B)
3. SF changes sign under the load → BM maximum there = $R_A \times 2 =$ **40 kN·m** (check: Wab/L = 30 × 2 × 4/6 = 40 ✔)

3.3 Bending stress — what happens inside the beam

When a beam bends in sagging, its top layers are **compressed** and bottom layers **stretched** (tension). In between there is a layer whose length neither increases nor decreases — the **neutral layer**; its line on the cross-section = the **neutral axis**, passing through the **centroid** of the section. Stress is zero at the neutral axis and increases in proportion to the distance from it — maximum at the outermost layer.

Derivation: Bending equation

1. Strain of a layer at distance y from the neutral axis $e = y/R$ (R = radius of curvature) → $\sigma = E\, y/R$
2. Internal resisting moment $M = \int \sigma\, y\, dA = (E/R) \int y^2\, dA = (E/R)\, I$
3. **$M/I = \sigma/y = E/R$**
4. **Section modulus $Z = I/y_{max}$ → $\sigma_{max} = M/Z$**. Rectangle: $Z = bd^2/6$; circle: $Z = \pi d^3/32$

Why an I-section? Stress is highest at the outer layers and nearly zero near the neutral axis. So keep the material outside (flanges) and less in the middle (web) — **more Z for less weight**. That is why rails and girders are I-shaped. Keeping a rectangular beam **on edge** (larger d) increases Z greatly, since $Z \propto d^2$.

Shear stress distribution: $\tau = F\, A\, \bar{y}/(I\, b)$. Parabolic in a rectangular section, maximum at the neutral axis = **1.5 × average**; in a circular section maximum = **4/3 × average**. Note — bending stress is maximum at the outer layer, shear stress maximum at the neutral axis.

Example: A 100 mm × 200 mm (b × d) rectangular beam has maximum BM = 20 kN·m. Maximum bending stress?

Solution:

1. $Z = bd^2/6 = 100 \times 200^2/6 = 6.667 \times 10^5$ mm³
2. $\sigma = M/Z = 20 \times 10^6/6.667 \times 10^5 =$ **30 N/mm²**
3. If laid flat (b = 200, d = 100): Z halves → $\sigma =$ **60 N/mm²**, double!

1. Key Points

- **Types of beams:** cantilever (one end fixed), simply supported, overhanging, fixed and continuous beams.
- **Types of loads:** point load, uniformly distributed load (UDL, w N/m) and uniformly varying load (UVL, triangular).
- **Sign convention:** upward force on the left gives positive SF; sagging BM (concave up) is positive and hogging is negative.
- **Relation of load, SF and BM:** $dF/dx = -w$ and $dM/dx = F$. Hence BM is maximum where SF is zero or changes sign.
- **Shape of diagrams:** point load → SFD rectangular (horizontal), BMD linear; UDL → SFD linear, BMD parabolic; UVL → SFD parabolic, BMD cubic.
- **Point of contraflexure:** the point where BM changes sign (BM = 0); it occurs in overhanging beams.

- **In a cantilever the maximum BM** is always at the fixed end; in a simply supported beam it is usually near mid-span.
- **Overhanging beam:** one or both ends extend beyond the supports; it has both positive and negative BM and a point of contraflexure.
- **SS beam with UVL (0 to w):** $R_A = wL/6$, $R_B = wL/3$; $M_{max} = wL^2/(9\sqrt{3}) \approx 0.064wL^2$ at $x = L/\sqrt{3}$.
- **Bending equation:** $M/I = \sigma/y = E/R$; section modulus $Z = I/y_{max}$; rectangle $Z = bd^2/6$, circle $Z = \pi d^3/32$.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Cantilever – W at free end	**$SF_{max} = W$, $M_{max} = WL$**
Cantilever – UDL over span	**$SF_{max} = wL$, $M_{max} = wL^2/2$**
SS beam – central W	**$R = W/2$, $M_{max} = WL/4$**
SS beam – UDL	**$R = wL/2$, $M_{max} = wL^2/8$**
SS beam – W at a, b	**$M_{max} = Wab/L$**
Cantilever – UVL (0 to w)	**$M_{max} = wL^2/6$**
Relations	**$dM/dx = F$, $dF/dx = -w$**

3. Detailed Explanation

- **Drawing the SFD:** first find reactions from $\Sigma Fy = 0$ and $\Sigma M = 0$; then move from left to right showing a jump in SF at each load.
- **Drawing the BMD:** BM at a section = moment of forces on one side. The slope of the BMD equals the SF, so BM is obtained from the area of the SFD.
- **Couple (moment) load:** a couple at a point causes a sudden jump in the BMD but has no effect on the SFD.
- **Pure bending:** the portion of a beam where SF = 0 and BM is constant, e.g. between two symmetrical point loads.
- **Neutral axis:** the layer with zero bending stress; it passes through the centroid. Compression above and tension below in sagging.
- **Shear stress in beams:** $\tau = FA\bar{y}/(Ib)$; for a rectangle $\tau_{max} = 1.5\tau_{avg}$ and for a circle $4/3\ \tau_{avg}$.
- **Beam of uniform strength:** a beam in which the maximum bending stress is the same at every section, e.g. a leaf spring.

4. Comparison Table — Cantilever vs Simply Supported

Basis	Cantilever	Simply supported
Supports	One end fixed	Both ends simply supported
BM_{max} (point load)	WL (at fixed end)	WL/4 (at centre)
BM_{max} (UDL)	$wL^2/2$	$wL^2/8$
Nature of BM	Hogging (negative)	Sagging (positive)

5. Solved Numerical Problems

Example 1. A 4 m simply supported beam carries a UDL of 10 kN/m. Find the reactions and maximum BM.

Solution:

1. $R_A = R_B = wL/2 = 10 \times 4/2 =$ **20 kN**
2. $M_{max} = wL^2/8 = 10 \times 16/8 =$ **20 kN·m** (at centre)

Example 2. A 3 m cantilever carries 5 kN at its free end. Find maximum SF and BM.

Solution:

1. $SF_{max} =$ **5 kN**
2. $M_{max} = WL = 5 \times 3 =$ **15 kN·m** (at the fixed end)

Example 3. A 6 m SS beam has 30 kN at 2 m from A. Find the maximum BM.

Solution:

1. $M_{max} = Wab/L = 30 \times 2 \times 4/6 =$ **40 kN·m**

Example 4. A 2 m cantilever carries 5 kN/m UDL and 10 kN at the free end. Find maximum SF and BM.

Solution:

1. $SF_{max} = 10 + 5 \times 2 =$ **20 kN**
2. $M_{max} = 10 \times 2 + 5 \times 2^2/2 =$ **30 kN·m**

Example 5. A 200 × 300 mm rectangular beam has M = 30 kN·m. Find the maximum bending stress.

Solution:

1. $Z = bd^2/6 = 200 \times 300^2/6 = 3 \times 10^6\ mm^3$
2. $\sigma = M/Z = 30 \times 10^6/3 \times 10^6 =$ **10 N/mm²**

6. Frequently Asked Facts

- SS beam with UDL: $M_{max} = wL^2/8$; central point load: WL/4.
- BM is maximum where SF = 0.
- Under UDL the SFD is linear and BMD parabolic.

7. Note (Confusing Facts)

- A point load causes a jump in the SFD but only a change of slope in the BMD.
- A couple load does not affect the SFD.
- BM in a cantilever is negative (hogging).

Remember: One line: dM/dx = F, dF/dx = −w. Assertion–Reason: (A) An overhanging beam has a point of contraflexure. (R) BM changes sign there. — Both true; R explains A.

30 Most Important One-Liner Facts — Shear Force & Bending Moment Diagrams

1. Beam with one end fixed?

— Cantilever.

2. UDL stands for?

— Uniformly Distributed Load.

3. Sign of sagging BM?

— Positive.

4. Sign of hogging BM?

— Negative.

5. dM/dx equals?

— Shear force.

6. Where is BM maximum?

— Where SF = 0.

7. SFD under point load?

— Rectangular (horizontal).

8. BMD under point load?

— Linear (triangular).

9. SFD under UDL?

— Linear (inclined).

10. BMD under UDL?

— Parabolic.

11. BMD under UVL?

— Cubic.

12. Cantilever, W at end — M_{max}?

— WL.

13. Cantilever, UDL — M_{max}?

— $wL^2/2$.

14. SS, central W — M_{max}?

— WL/4.

15. SS, UDL — M_{max}?

— $wL^2/8$.

16. Point where BM changes sign?

— Point of contraflexure.

17. Effect of couple on SFD?

— None.

18. Location of M_{max} in cantilever?

— At fixed end.

19. SF in pure bending?

— Zero.

20. SS beam, UDL — reaction?

— wL/2.

21. Bending equation?

— $M/I = \sigma/y = E/R$.

22. Section modulus of rectangle?

— $bd^2/6$.

23. Section modulus of circle?

— $\pi d^3/32$.

24. Bending stress at neutral axis?

— Zero.

25. Neutral axis passes through?

— Centroid.

26. τ_{max}/τ_{avg} for rectangle?

— 1.5.

27. τ_{max}/τ_{avg} for circle?

— 4/3.

28. Example of uniform strength beam?

— Leaf spring.

29. Special point in overhanging beam?

— Point of contraflexure.

30. SS beam, UVL — M_{max}?

— $wL^2/(9\sqrt{3})$.

Practice MCQs — Shear Force & Bending Moment Diagrams

Q1. For an SS beam with UDL over the whole span, M_{max} =

(a) $wL^2/2$

(b) $wL^2/4$

(c) $wL^2/8$

(d) $wL^2/12$

✔ **Answer: (c) $wL^2/8$**

Explanation: $wL^2/8$ at the centre.

Q2. For a cantilever with W at the free end, M_{max} =

(a) WL/4

(b) WL/2

(c) WL

(d) 2WL

✔ **Answer: (c) WL**

Explanation: WL at the fixed end.

Q3. Bending moment is maximum where:

(a) SF is maximum

(b) SF is zero

(c) Load is maximum

(d) Deflection is zero

✔ **Answer: (b) SF is zero**

Explanation: dM/dx = F = 0.

Q4. Shape of BMD under UDL:

(a) Rectangular

(b) Triangular

(c) Parabolic

(d) Cubic

✔ **Answer: (c) Parabolic**

Explanation: $M \propto x^2$.

Q5. SFD under point loads is:

(a) Horizontal lines

(b) Inclined lines

(c) Parabola

(d) Circle

✔ **Answer: (a) Horizontal lines**

Explanation: SF is constant between loads.

Q6. At a point of contraflexure:

(a) SF = 0

(b) BM = 0

(c) BM is maximum

(d) Deflection is maximum

✔ **Answer: (b) BM = 0**

Explanation: BM changes sign.

Q7. For a cantilever with UDL, M_{max} =

(a) $wL^2/2$ (b) $wL^2/8$

(c) $wL^2/6$ (d) wL

✔ **Answer: (a) $wL^2/2$**

Explanation: $wL^2/2$ at the fixed end.

Q8. dM/dx equals:

(a) Load intensity (b) Shear force

(c) Deflection (d) Slope

✔ **Answer: (b) Shear force**

Explanation: Slope of BM = SF.

Q9. A couple applied at a point affects:

(a) Only SFD (b) Only BMD

(c) Both (d) Neither

✔ **Answer: (b) Only BMD**

Explanation: Jump in BMD; SFD unaffected.

Q10. 6 m SS beam with 20 kN at centre — M_{max}:

(a) 20 kN·m (b) 30 kN·m

(c) 60 kN·m (d) 120 kN·m

✔ **Answer: (b) 30 kN·m**

Explanation: $WL/4 = 20 \times 6/4 = 30$.

Q11. The bending equation is:

(a) $M/I = \sigma/y = E/R$ (b) $T/J = \tau/R = G\theta/L$

(c) $M/Z = E/R$ (d) $\sigma = P/A$

✔ **Answer: (a) $M/I = \sigma/y = E/R$**

Explanation: Standard.

Q12. Section modulus of a rectangular section:

(a) $bd^3/12$ (b) $bd^2/6$

(c) $bd^2/12$ (d) $bd/6$

✔ **Answer: (b) $bd^2/6$**

Explanation: $I/y = (bd^3/12)/(d/2)$.

Q13. Bending stress at the neutral axis:

(a) Maximum (b) Zero

(c) Average (d) Infinite

✔ **Answer: (b) Zero**

Explanation: y = 0.

Q14. Max shear stress in a rectangular beam is ____ the average:

(a) 1.0 (b) 4/3

(c) 1.5 (d) 2.0

✔ **Answer: (c) 1.5**

Explanation: 1.5 times.

Q15. Cantilever 2 m, UDL 5 kN/m + 10 kN at end — M_{max}:

(a) 20 kN·m (b) 30 kN·m

(c) 40 kN·m (d) 10 kN·m

✔ **Answer: (b) 30 kN·m**

Explanation: 20 + 10 = 30.

SSC JE / UPSSSC JE Previous Year Questions — Shear Force & Bending Moment Diagrams

Q1. Maximum BM of an SS beam with central point load W: SSC JE (Previous Year)

(a) WL/2 (b) WL/4

(c) WL/8 (d) WL

✔ **Answer: (b) WL/4**

Explanation: M = WL/4.

Q2. A portion of a beam with zero SF and constant BM is in: SSC JE (Previous Year)

(a) Pure bending (b) Pure shear

(c) Torsion (d) Buckling

✔ **Answer: (a) Pure bending**

Explanation: Pure bending.

Q3. Under UVL the SFD is: SSC JE (Previous Year)

(a) Linear (b) Parabolic

(c) Cubic (d) Rectangular

✔ **Answer: (b) Parabolic**

Explanation: SF ∝ x^2.

Q4. Maximum BM in a cantilever occurs at: SSC JE (Previous Year)

(a) Free end (b) Middle

(c) Fixed end (d) Anywhere

✔ **Answer: (c) Fixed end**

Explanation: At the fixed end.

Q5. Maximum BM of an SS beam under UDL: SSC JE (Previous Year)

(a) $wL^2/2$ (b) $wL^2/8$

(c) $wL^2/12$ (d) $wL^2/4$

✔ **Answer: (b) $wL^2/8$**

Explanation: $wL^2/8$.

Q6. Under sagging BM the top fibres of a beam are in: SSC JE (Previous Year)

(a) Tension (b) Compression

(c) Shear (d) Zero stress

✔ **Answer: (b) Compression**

Explanation: Compression at top, tension at bottom.

UPSSSC ALERT

Chapter 4: Composite Bars & Thermal Stresses

Composite Bars & Thermal Stresses – Basic Concept

When bars of two or more materials are joined so that all deform equally under load, the arrangement is a **composite bar**. If the expansion of a bar due to temperature change is prevented, **thermal stress** develops in it.

Understand in Detail — Composite Bars & Thermal Stresses

4.1 Composite bars — when two materials share a load

In an RCC column, concrete and steel bars carry the load together; concrete filled in a pipe is similar. A **composite bar** is made of two or more materials in which, under load, **all parts stretch/compress equally** (because they are bonded together).

Two basic conditions — the key to this whole chapter: (1) **equilibrium** — total load = sum of loads carried by both parts: $P = \sigma_1 A_1 + \sigma_2 A_2$; (2) **compatibility** — both have equal elongation (and, for equal lengths, equal strain): $\sigma_1/E_1 = \sigma_2/E_2$.

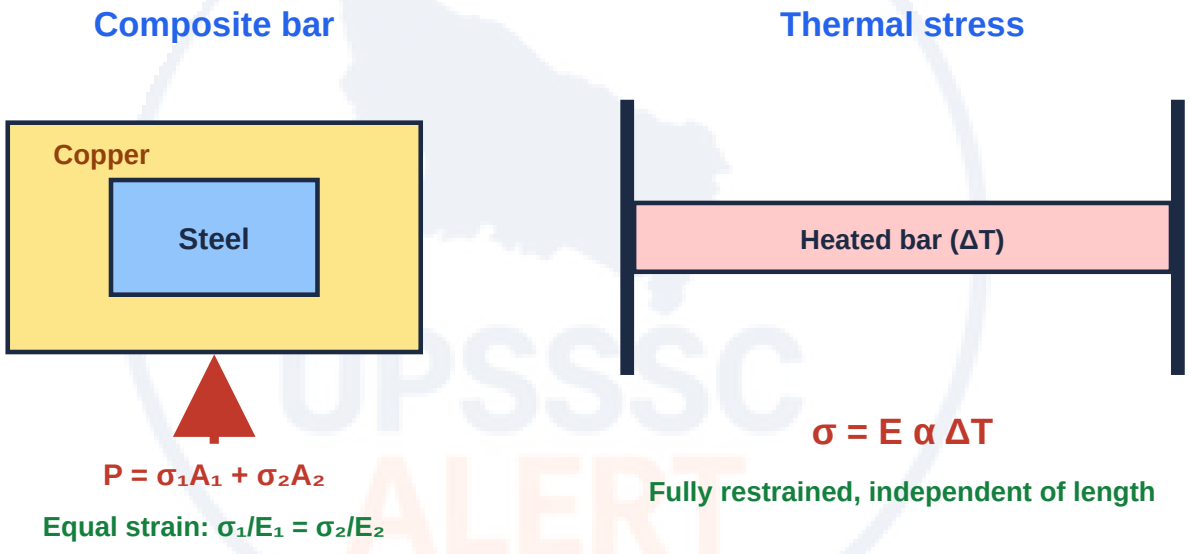


Fig 3.4: Composite bar and thermal stress in a fully restrained bar

Derivation: Stresses in a composite bar

1. From compatibility $\sigma_1 = (E_1/E_2)\,\sigma_2$; **modular ratio $m = E_1/E_2$** → $\sigma_1 = m\,\sigma_2$
2. Substitute in equilibrium: $P = \sigma_2(m\,A_1 + A_2)$ → **$\sigma_2 = P/(A_2 + mA_1)$** ; $\sigma_1 = m\,\sigma_2$
3. **Meaning:** the material with higher E (steel) takes more stress — in steel–concrete m ≈ 15–18, i.e. steel stress is 15 times the concrete stress
4. **Equivalent area** = $A_2 + mA_1$ — as if the whole bar were made of material 2 alone

Example: A 300 mm × 300 mm concrete column has 4 steel bars (20 mm dia), load 1000 kN, m = 15. Stresses in both?

Solution:

1. $A_s = 4 \times (\pi/4)(20)^2 = 1256.6$ mm² ; $A_c = 90000 - 1256.6 = 88743.4$ mm²
2. Equivalent area = 88743.4 + 15 × 1256.6 = 107592.4 mm²
3. $\sigma_c = 10^6/107592.4 =$ **9.29 N/mm²** ; $\sigma_s = 15 \times 9.29 =$ **139.4 N/mm²**

4.2 Thermal stresses — when temperature 'pushes'

Why are gaps left between rail tracks in summer? Because metal expands when heated. If a bar is **free**, it simply expands $\delta L = L\,\alpha\,\Delta T$ — no stress (α = coefficient of linear expansion, steel ≈ 12×10^{-6} /°C). But if both ends are **held by rigid walls**, the walls won't let it expand — that expansion is 'suppressed' inside the bar as a **compressive stress**. This is **thermal stress**.

Derivation: Thermal stress

1. Prevented expansion = $L\,\alpha\,\Delta T$; thermal strain = prevented expansion/L = $\alpha\,\Delta T$
2. $\boldsymbol{\sigma = E\,\alpha\,\Delta T}$ — independent of length or area!
3. **Partial restraint:** if the wall yields by δ: $\sigma = E(L\,\alpha\,\Delta T - \delta)/L$
4. Temperature rise → compression, fall → tension (in a restrained bar)

Temperature on a composite bar: steel and copper bars bonded together are heated — copper has the higher α (17×10^{-6}) and wants to expand more, steel less. Being bonded they settle at a common length: **the one with higher α (copper) goes into compression, the lower-α one (steel) into tension**. Equations: $\sigma_s A_s = \sigma_c A_c$ (no external load) and $\sigma_s/E_s + \sigma_c/E_c = (\alpha_c - \alpha_s)\Delta T$.

Example: A steel bar between two rigid walls is heated by 30°C. E = 200 GPa, $\alpha = 12 \times 10^{-6}$/°C. Stress? What if the wall yields 0.2 mm and L = 5 m?

Solution:

1. $\sigma = E\,\alpha\,\Delta T = 200000 \times 12 \times 10^{-6} \times 30 =$ **72 N/mm² (compressive)**
2. Free expansion = $5000 \times 12 \times 10^{-6} \times 30 = 1.8$ mm ; prevented = 1.8 − 0.2 = 1.6 mm
3. $\sigma = 200000 \times 1.6/5000 =$ **64 N/mm²**

1. Key Points

- **Conditions for a composite bar:** (i) total load = sum of loads carried: $P = \sigma_1 A_1 + \sigma_2 A_2$; (ii) equal strain: $\sigma_1/E_1 = \sigma_2/E_2$.
- **Modular ratio:** $m = E_1/E_2$, so $\sigma_1 = m \cdot \sigma_2$. The material with higher E carries higher stress.
- **Thermal strain:** $\varepsilon_t = \alpha\Delta T$; free expansion $\delta = L\alpha\Delta T$.
- **Thermal stress (fully restrained):** $\sigma = E\alpha\Delta T$ — independent of length and cross-section. Heating gives compression; cooling gives tension.
- **Partial restraint:** if a gap δ_g is allowed, $\sigma = E(L\alpha\Delta T - \delta_g)/L$.

- **Thermal stress in composite bars:** on heating, the material with higher α (copper) goes into compression and the lower-α one (steel) into tension; force balance $\sigma_1 A_1 = \sigma_2 A_2$.
- **Coefficient of linear expansion (α):** steel ≈ 12×10^{-6} /°C, copper ≈ 17×10^{-6} /°C, aluminium ≈ 23×10^{-6} /°C.
- **Use of modular ratio:** a composite bar is converted to one equivalent material with $A_{eq} = A_1 + mA_2$.
- **Effect of α:** for the same ΔT the higher-α material wants to expand more; when restrained it develops more stress.
- **Pipes and bolts:** bolts of a hot pipe flange go into tension on cooling; long welded rails need stress control.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Load balance	$\mathbf{P = \sigma_1 A_1 + \sigma_2 A_2}$
Equal strain	$\mathbf{\sigma_1/E_1 = \sigma_2/E_2}$
Modular ratio	$\mathbf{m = E_1/E_2}$
Equivalent E	$\mathbf{E_{eq} = (E_1A_1 + E_2A_2)/(A_1 + A_2)}$
Free thermal expansion	$\mathbf{\delta = L\,\alpha\,\Delta T}$
Thermal stress (full restraint)	$\mathbf{\sigma = E\,\alpha\,\Delta T}$
Partial restraint	$\mathbf{\sigma = E(L\alpha\Delta T - \delta_g)/L}$
Composite bar (thermal)	$\mathbf{\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T}$

3. Detailed Explanation

- **Statically indeterminate:** in a composite bar the unknown forces cannot be found from equilibrium alone; the compatibility (deformation) condition is also needed.
- **RCC column:** concrete and steel bars share the load; stress in steel = m × stress in concrete, with m ≈ 13–18.
- **Examples:** gaps between rails, expansion joints in bridges and loops in steam pipes are provided to avoid thermal stresses.
- **Why independent of length:** stress = E × prevented strain = E × αΔT; L cancels in the strain.
- **Derivation for composite bars:** the difference in free expansion $(\alpha_1 - \alpha_2)\Delta T \cdot L$ equals the sum of elastic strains: $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$, with $\sigma_1 A_1 = \sigma_2 A_2$.
- **Bolt and tube system:** tightening the nut puts the bolt in tension and the tube in compression; equal forces: $\sigma_b A_b = \sigma_t A_t$.

4. Comparison Table — Heating vs Cooling (restrained bar)

Basis	Heating	Cooling
Tendency	Expand	Contract
Type of stress	Compressive	Tensile
Composite: higher-α material	Compression	Tension

5. Solved Numerical Problems

Example 1. A 3 m steel rod is fixed at both ends and heated by 50°C. E = 200 GPa, α = 12 × 10^{-6} /°C. Find the thermal stress.

Solution:

1. σ = EαΔT = 200000 × 12 × 10^{-6} × 50
2. σ = **120 N/mm²** (compressive); length has no effect

Example 2. If the above rod is given a gap of 0.6 mm, find the stress.

Solution:

1. Free expansion = 3000 × 12 × 10^{-6} × 50 = 1.8 mm; prevented = 1.8 − 0.6 = 1.2 mm
2. σ = E × 1.2/3000 = **80 N/mm²**

Example 3. A composite bar of steel (A = 1000 mm², E = 200 GPa) and copper (A = 2000 mm², E = 100 GPa) carries 400 kN compression. Find the stresses.

Solution:

1. $\sigma_s = (E_s/E_c)\sigma_c = 2\sigma_c$
2. 400000 = $2\sigma_c \times 1000 + \sigma_c \times 2000 = 4000\sigma_c$
3. σ_c = **100 N/mm²**, σ_s = **200 N/mm²**

Example 4. A steel (A = 500 mm²) – copper (A = 1000 mm²) composite bar is heated by 40°C. E_s = 200, E_c = 100 GPa; α_s = 12, α_c = 18 × 10^{-6} /°C. Find the stresses.

Solution:

1. Force balance: $\sigma_s \times 500 = \sigma_c \times 1000 \Rightarrow \sigma_s = 2\sigma_c$
2. $\sigma_s/200000 + \sigma_c/100000 = 6 \times 10^{-6} \times 40 = 2.4 \times 10^{-4}$
3. $2\sigma_c/200000 + \sigma_c/100000 = 2\sigma_c/100000 = 2.4 \times 10^{-4}$
4. σ_c = **12 N/mm²** (compression), σ_s = **24 N/mm²** (tension)

6. Frequently Asked Facts

- Thermal stress σ = EαΔT is independent of length and area.
- In a composite bar strain is equal; stresses are in the ratio of E.
- A restrained bar heated develops compressive stress.

7. Note (Confusing Facts)

- In a composite bar, strains (not stresses) are equal.
- Thermal stress arises only when expansion is prevented; a free bar has zero stress.
- On heating a composite bar, the lower-α material goes into tension.

Remember: One line: $\sigma_1/\sigma_2 = E_1/E_2$ (equal strain). Assertion–Reason: (A) Gaps are left between rails. (R) Prevented expansion in summer causes compressive stress. — Both true; R explains A.

30 Most Important One-Liner Facts — Composite Bars & Thermal Stresses

1. What is equal in a composite bar?

— Strain.

2. Load in a composite bar?

— $P = \sigma_1 A_1 + \sigma_2 A_2$.

3. Modular ratio?

— E_1/E_2.

4. Which carries more stress?

— Material with higher E.

5. Thermal strain?

— $\alpha\Delta T$.

6. Free expansion?

— $L\alpha\Delta T$.

7. Thermal stress?

— $E\alpha\Delta T$.

8. Depends on length?

— No.

9. Heated restrained bar?

— Compressive stress.

10. Cooled restrained bar?

— Tensile stress.

11. Thermal stress in a free bar?

— Zero.

12. α of steel?

— $\approx 12 \times 10^{-6}$ /°C.

13. α of copper?

— $\approx 17 \times 10^{-6}$ /°C.

14. Composite bar is which type of problem?

— Statically indeterminate.

15. Stress in steel in RCC?

— m × stress in concrete.

16. Why gaps in rails?

— To avoid thermal stress.

17. Higher-α material on heating?

— In compression.

18. Stress with partial restraint?

— $E(L\alpha\Delta T - \delta)/L$.

19. Equivalent E?

— $(E_1A_1 + E_2A_2)/(A_1 + A_2)$.

20. α of aluminium?

— $\approx 23 \times 10^{-6}$ /°C.

21. Equivalent area?

— $A_1 + mA_2$.

22. Composite thermal equation?

— $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$.

23. Force balance (thermal)?

— $\sigma_1 A_1 = \sigma_2 A_2$.

24. Bolt in bolt–tube system?

— Tension.

25. Tube in bolt–tube system?

— Compression.

26. Flange bolts when hot pipe cools?

— Tension.

27. Modular ratio in RCC?

— ≈ 13–18.

28. Net force in composite bar (thermal, no load)?

— Zero.

29. Thermal stress independent of?

— Length and area.

30. Why expansion joints in bridges?

— To avoid thermal stress.

Practice MCQs — Composite Bars & Thermal Stresses

Q1. Thermal stress in a fully restrained bar is:

(a) EαΔT
(b) LαΔT
(c) EαΔT/L
(d) αΔT/E

✔ **Answer: (a) EαΔT**

Explanation: σ = E × αΔT.

Q2. Thermal stress does not depend on:

(a) E
(b) α
(c) Length
(d) ΔT

✔ **Answer: (c) Length**

Explanation: L cancels out.

Q3. In a composite bar both materials have equal:

(a) Stress
(b) Strain
(c) Load
(d) Area

✔ **Answer: (b) Strain**

Explanation: Both deform together.

Q4. On heating a restrained bar, the stress is:

(a) Tensile
(b) Compressive
(c) Shear
(d) Zero

✔ **Answer: (b) Compressive**

Explanation: Prevented expansion pushes back.

Q5. Modular ratio is:

(a) E_1/E_2
(b) A_1/A_2
(c) $\sigma_1 \times \sigma_2$
(d) α_1/α_2

✔ **Answer: (a) E_1/E_2**

Explanation: $m = E_1/E_2$.

Q6. In a steel (200 GPa) – copper (100 GPa) composite bar, σ_s/σ_c =

(a) 0.5
(b) 1
(c) 2
(d) 4

✔ **Answer: (c) 2**

Explanation: Equal strain gives stress ratio = E ratio.

Q7. Thermal stress in a free bar on heating:

(a) Compressive
(b) Tensile
(c) Zero
(d) Shear

✔ **Answer: (c) Zero**

Explanation: Expansion is free.

Q8. On heating a composite bar, the lower-α material is in:

(a) Compression
(b) Tension
(c) Zero stress
(d) Shear

✔ **Answer: (b) Tension**

Explanation: The higher-α material pulls it.

Q9. E = 200 GPa, α = 12 × 10^{-6}, ΔT = 100°C — thermal stress:

(a) 120 MPa
(b) 240 MPa
(c) 24 MPa
(d) 2400 MPa

✔ **Answer: (b) 240 MPa**

Explanation: 200000 × 12e-6 × 100 = 240.

Q10. A composite bar problem is:

(a) Statically determinate
(b) Statically indeterminate
(c) Dynamic
(d) Unstable

✔ **Answer: (b) Statically indeterminate**

Explanation: Compatibility condition needed.

Q11. A composite bar under temperature change with no external load:

(a) Both in tension
(b) One in tension, other in compression
(c) Both in compression
(d) No stress

✔ **Answer: (b) One in tension, other in compression**

Explanation: Internal forces balance.

Q12. Tightening the nut in a bolt–tube system puts the tube in:

(a) Tension
(b) Compression
(c) Torsion
(d) Nothing

✔ **Answer: (b) Compression**

Explanation: Bolt in tension, tube in compression.

Q13. Equivalent area of a composite bar (in material 1):

(a) $A_1 + A_2$
(b) $A_1 + mA_2$
(c) $mA_1 + A_2$
(d) A_1A_2

✔ **Answer: (b) $A_1 + mA_2$**

Explanation: A_2 is transformed by $m = E_2/E_1$.

Q14. Correct equation for a composite bar on heating:

(a) $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$ (b) $\sigma_1 E_1 = \sigma_2 E_2$

(c) $\sigma_1 + \sigma_2 = E\alpha\Delta T$ (d) $\sigma_1/\sigma_2 = \alpha_1/\alpha_2$

✔ **Answer: (a) $\sigma_1/E_1 + \sigma_2/E_2 = (\alpha_1 - \alpha_2)\Delta T$**

Explanation: Sum of strains = difference in expansion.

Q15. Full restraint, E = 100 GPa, $\alpha = 20 \times 10^{-6}$, $\Delta T = 50$ — stress:

(a) 50 MPa (b) 100 MPa

(c) 200 MPa (d) 10 MPa

✔ **Answer: (b) 100 MPa**

Explanation: 100000 × 20e-6 × 50 = 100.

SSC JE / UPSSSC JE Previous Year Questions — Composite Bars & Thermal Stresses

Q1. A bar fixed at both ends is heated. The stress induced is:

SSC JE (Previous Year)

(a) Tensile (b) Compressive

(c) Shear (d) Zero

✔ **Answer: (b) Compressive**

Explanation: Prevented expansion gives compression.

Q2. Thermal stress is given by:

SSC JE (Previous Year)

(a) $\alpha\Delta T/E$ (b) $E\alpha\Delta T$

(c) $E\alpha\Delta T/L$ (d) $\alpha E/\Delta T$

✔ **Answer: (b) $E\alpha\Delta T$**

Explanation: $\sigma = E\alpha\Delta T$.

Q3. In a composite bar, the material with higher E:

SSC JE (Previous Year)

(a) Takes less stress (b) Takes more stress

(c) Takes equal stress (d) Takes no load

✔ **Answer: (b) Takes more stress**

Explanation: $\sigma \propto E$.

Q4. A bar fixed at both ends when cooled develops:

SSC JE (Previous Year)

(a) Compressive stress (b) Tensile stress

(c) Shear (d) Nothing

✔ **Answer: (b) Tensile stress**

Explanation: Contraction is prevented.

Q5. Thermal strain is:

SSC JE (Previous Year)

(a) αΔT

(b) EαΔT

(c) LαΔT

(d) αΔT/E

✔ **Answer: (a) αΔT**

Explanation: ε = αΔT.

UPSSSC ALERT

Chapter 5: Torsion of Circular Shafts

Torsion of Circular Shafts – Basic Concept

When a torque acts on a shaft about its axis, the shaft twists and **shear stress** develops in its cross-section. All power-transmitting shafts are subjected to torsion.

Understand in Detail — Torsion of Circular Shafts

5.1 Torsion — what happens when a shaft is twisted

A shaft carrying power from a motor to a pump, a car's propeller shaft, a screwdriver — all are twisted by a **torque**. Draw a straight line along a rubber rod and twist it — the line becomes a **helix**. This means every layer of the rod is sliding over its neighbour — i.e. **shear stress** is produced.

Assumptions of the theory: material homogeneous and elastic; shaft circular; plane sections remain plane and radii remain straight after twisting; torque constant. **Angle of twist (θ)** — rotation of one end relative to the other (radians).

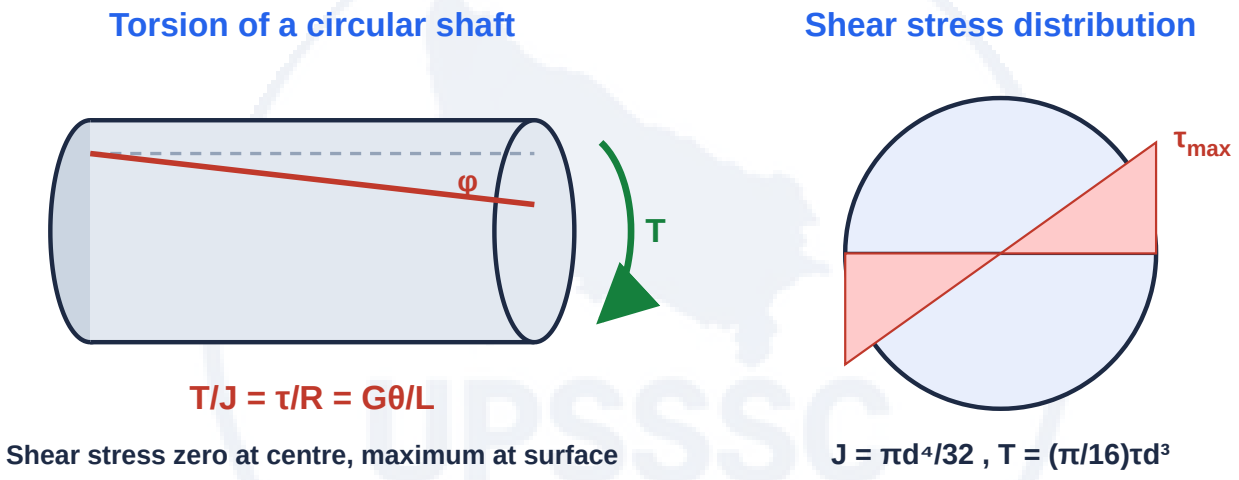


Fig 3.5: Torsion of a circular shaft and shear stress distribution

Derivation: Torsion equation

1. Shear strain at outer surface $\varphi = R\theta/L$ (arc = $R\theta = L\varphi$) ; at radius r: $\varphi = r\theta/L$
2. $\tau = G\varphi = G\, r\, \theta/L \rightarrow$ **$\tau/r = G\theta/L$** — zero at centre, maximum at surface (linear)
3. Resisting torque $T = \int \tau\, r\, dA = (G\theta/L) \int r^2\, dA = (G\theta/L)\, J$
4. **$T/J = \tau/R = G\theta/L$** ; solid shaft $J = \pi D^4/32$; hollow $J = \pi(D^4 - d^4)/32$
5. Solid shaft: **$T = (\pi/16)\, \tau\, D^3$** ; hollow: $T = (\pi/16)\, \tau\, (D^4 - d^4)/D$
6. **Power $P = 2\pi NT/60$** (W, T in N·m) ; **torsional rigidity = GJ**

Why is a hollow shaft more efficient? Near the centre the stress is nearly zero — that material is almost useless. Remove it and place it outside to get **more torque for the same weight**. That is why car

propeller shafts and bicycle frames are made of hollow tubes. **Shafts in series** (one after another): same torque, angles of twist add; **in parallel (compound)**: same angle of twist, torque is shared.

Example: A solid shaft transmits 100 kW at 200 rpm. Allowable $\tau = 60$ N/mm². Diameter?
Solution:

1. $T = 60P/(2\pi N) = 60 \times 100000/(2\pi \times 200) = 4775$ N·m $= 4.775 \times 10^6$ N·mm
2. $D^3 = 16T/(\pi\,\tau) = 16 \times 4.775 \times 10^6/(\pi \times 60) = 405284$
3. D = **74 mm** (take standard 75 mm)

Note: A **close-coiled helical spring** is also an example of torsion — an axial load W puts a torque WR on the spring wire: $\tau = 16WR/(\pi d^3)$; deflection $\delta = 64WR^3n/(Gd^4)$; stiffness $k = W/\delta = Gd^4/(64R^3n)$. Springs in series: $1/k = 1/k_1 + 1/k_2$; in parallel: $k = k_1 + k_2$.

1. Key Points

- **Torsion equation:** $T/J = \tau/R = G\theta/L$, where T is torque, J polar moment of inertia, τ surface shear stress, R radius, G modulus of rigidity, θ angle of twist (radian), L length.
- **Polar moment of inertia:** solid shaft $J = \pi d^4/32$; hollow shaft $J = \pi(D^4 - d^4)/32$.
- **Torque capacity:** solid shaft $T = (\pi/16)\tau d^3$; hollow shaft $T = (\pi/16)\tau(D^4 - d^4)/D$.
- **Polar section modulus (Z_p):** J/R; $\pi d^3/16$ for a solid shaft. It measures torsional strength.
- **Power:** $P = 2\pi NT/60$ (W), N in rpm.
- **Stress distribution:** shear stress is zero at the centre and increases linearly with radius to a maximum at the surface.
- **Torsional rigidity:** GJ, the torque per unit twist per unit length. GJ/L is the torsional stiffness.
- **Hollow shaft is better:** for the same weight a hollow shaft is stronger and stiffer than a solid one, as material near the centre carries little stress.
- **Diameter for power:** find T from $P = 2\pi NT/60$, then $d = (16T/\pi\tau)^{1/3}$.
- **Hollow vs solid (same weight):** with inner diameter = ½ outer, a hollow shaft carries about 1.44 times the torque of a solid shaft of equal weight.
- **Combined bending and torsion:** equivalent torque $T_e = \sqrt{(M^2 + T^2)}$ and equivalent BM $M_e = ½[M + \sqrt{(M^2 + T^2)}]$.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Torsion equation	**$T/J = \tau/R = G\theta/L$**
J (solid)	**$\pi d^4/32$**
J (hollow)	**$\pi(D^4 - d^4)/32$**
T (solid)	**$(\pi/16)\,\tau\, d^3$**
T (hollow)	**$(\pi/16)\,\tau\,(D^4 - d^4)/D$**
Power	**$P = 2\pi NT / 60$**
Angle of twist	**$\theta = TL / GJ$**
Shafts in series	**Same T , $\theta = \theta_1 + \theta_2$**
Shafts in parallel	**Same θ , $T = T_1 + T_2$**

3. Detailed Explanation

- **Assumptions:** material homogeneous and isotropic; plane sections remain plane; radii remain straight; stress within the elastic limit.
- **Series and parallel:** shafts in series carry the same torque and twists add; in parallel the twist is the same and torque divides.
- **Shaft design:** diameter is found from both strength ($\tau \le$ allowable) and stiffness ($\theta \le$ allowable, e.g. 1°/m), and the larger value is taken.
- **Doubling the diameter:** torque capacity becomes 8 times (d^3) and stiffness 16 times (d^4).
- **Close-coiled helical spring:** torsional stress in wire $\tau = 16WR/(\pi d^3)$; deflection $\delta = 64WR^3n/(Gd^4)$; stiffness $k = Gd^4/(64R^3n)$.
- **Spring combinations:** series $1/k = 1/k_1 + 1/k_2$; parallel $k = k_1 + k_2$.

4. Comparison Table — Solid vs Hollow Shaft

Basis	Solid	Hollow
J	$\pi d^4/32$	$\pi(D^4 - d^4)/32$
Strength for same weight	Less	More
Material at centre	Under-used	Removed
Cost/manufacture	Cheaper, simpler	Costlier

5. Solved Numerical Problems

Example 1. A 50 mm solid shaft has allowable shear stress 60 N/mm². Find the maximum torque.
Solution:

1. $T = (\pi/16) \times 60 \times 50^3 = (\pi/16) \times 60 \times 125000$
2. $T = 1.473 \times 10^6$ N·mm = **1.47 kN·m**

Example 2. A shaft transmits 1.5 kN·m at 200 rpm. Find the power.
Solution:

1. $P = 2\pi \times 200 \times 1500/60$ = **31.4 kW**

Example 3. For the shaft above (d = 50 mm, L = 2 m, G = 80 GPa) at 1.47 kN·m, find the angle of twist.
Solution:

1. $J = \pi \times 50^4/32 = 613{,}592$ mm⁴
2. $\theta = TL/GJ = 1.473 \times 10^6 \times 2000/(80000 \times 613592) = 0.060$ rad
3. θ = **3.44°**

Example 4. A solid shaft transmits 30 kW at 150 rpm; τ = 60 N/mm². Find the diameter.
Solution:

1. $T = 30000 \times 60/(2\pi \times 150) = 1910$ N·m $= 1.91 \times 10^6$ N·mm
2. $d^3 = 16T/(\pi\tau) = 16 \times 1.91 \times 10^6/(\pi \times 60) = 162{,}100$
3. d = **54.5 mm ≈ 55 mm**

Example 5. A shaft has M = 3 kN·m and T = 4 kN·m. Find equivalent torque and BM.
Solution:

1. $T_e = \sqrt{(9 + 16)}$ = **5 kN·m**
2. $M_e = \frac{1}{2}(3 + 5)$ = **4 kN·m**

6. Frequently Asked Facts

- $T/J = \tau/R = G\theta/L$ — the whole chapter is built on this.
- Solid shaft: $T = (\pi/16)\tau d^3$; $P = 2\pi NT/60$.
- A hollow shaft is stronger for the same weight.

7. Note (Confusing Facts)

- Shear stress is zero at the centre and maximum at the surface — not the reverse.
- θ is in radians; multiply by $180/\pi$ for degrees.
- Diameter doubled → strength 8 times, stiffness 16 times.

Remember: One line: GJ = torsional rigidity; J/R = polar section modulus. Assertion–Reason: (A) A hollow shaft is stronger than a solid one of the same weight. (R) Material near the centre is lightly

stressed. — Both true; R explains A.

30 Most Important One-Liner Facts — Torsion of Circular Shafts

1. Torsion equation?

— $T/J = \tau/R = G\theta/L$.

2. J of solid shaft?

— $\pi d^4/32$.

3. J of hollow shaft?

— $\pi(D^4 - d^4)/32$.

4. Torque of solid shaft?

— $(\pi/16)\tau d^3$.

5. Power formula?

— $2\pi NT/60$.

6. Angle of twist?

— TL/GJ.

7. Torsional rigidity?

— GJ.

8. Polar section modulus (solid)?

— $\pi d^3/16$.

9. Shear stress at centre?

— Zero.

10. Maximum shear stress at?

— Outer surface.

11. Stress distribution?

— Linear with radius.

12. Diameter doubled — strength?

— 8 times.

13. Diameter doubled — stiffness?

— 16 times.

14. Same in series shafts?

— Torque.

15. Same in parallel shafts?

— Angle of twist.

16. Better shaft for same weight?

— Hollow.

17. Unit of θ in formula?

— Radian.

18. Type of stress in torsion?

— Shear.

19. G in torsion?

— Modulus of rigidity.

20. Two design criteria for shafts?

— Strength and stiffness.

21. Diameter from power?

— $(16T/\pi\tau)^{1/3}$.

22. Equivalent torque?

— $\sqrt{(M^2 + T^2)}$.

23. Equivalent BM?

— $\frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$.

24. Stress in spring wire?

— $16WR/\pi d^3$.

25. Spring deflection?

— $64WR^3n/Gd^4$.

26. Series spring stiffness?

— $1/k = 1/k_1 + 1/k_2$.

27. Parallel spring stiffness?

— $k_1 + k_2$.

28. Stress in close-coiled spring?

— Torsional (shear).

29. 1 kW equals?

— 1000 W.

30. G of steel?

— ≈ 80 GPa.

Practice MCQs — Torsion of Circular Shafts

Q1. The torsion equation is:

(a) $T/J = \tau/R = G\theta/L$

(b) $M/I = \sigma/y = E/R$

(c) $T/I = \tau/R = E\theta/L$

(d) $P/A = \sigma$

✔ **Answer: (a) $T/J = \tau/R = G\theta/L$**

Explanation: Torsion equation.

Q2. Polar moment of inertia of a solid shaft:

(a) $\pi d^4/64$

(b) $\pi d^4/32$

(c) $\pi d^3/16$

(d) $\pi d^3/32$

✔ **Answer: (b) $\pi d^4/32$**

Explanation: $J = 2I = \pi d^4/32$.

Q3. Torque capacity of a solid shaft:

(a) $(\pi/16)\tau d^3$

(b) $(\pi/32)\tau d^3$

(c) $(\pi/16)\tau d^4$

(d) $(\pi/32)\tau d^4$

✔ **Answer: (a) $(\pi/16)\tau d^3$**

Explanation: $T = \tau \times J/R$.

Q4. Shear stress at the centre of a shaft:

(a) Maximum

(b) Zero

(c) Average

(d) Infinite

✔ **Answer: (b) Zero**

Explanation: $\tau \propto r$.

Q5. Doubling the diameter, the strength of a shaft becomes:

(a) 2 times

(b) 4 times

(c) 8 times

(d) 16 times

✔ **Answer: (c) 8 times**

Explanation: $T \propto d^3$.

Q6. Torsional rigidity is:

(a) GJ

(b) EI

(c) GA

(d) EJ

✔ **Answer: (a) GJ**

Explanation: GJ.

Q7. Power transmitted by a shaft:

(a) $2\pi NT/60$ (b) $\pi NT/60$

(c) $2\pi NT$ (d) $NT/60$

✔ **Answer: (a) $2\pi NT/60$**

Explanation: $P = T\omega$.

Q8. In shafts connected in series, the same is:

(a) Angle of twist (b) Torque

(c) Diameter (d) Stress

✔ **Answer: (b) Torque**

Explanation: The same torque passes through both.

Q9. A hollow shaft compared to a solid one of same weight is:

(a) Weaker (b) Stronger

(c) Equal (d) Cannot say

✔ **Answer: (b) Stronger**

Explanation: Material is used where stress is high.

Q10. Stress produced in torsion:

(a) Tensile (b) Compressive

(c) Shear (d) Bending

✔ **Answer: (c) Shear**

Explanation: Shear stress.

Q11. Equivalent torque under combined bending and torsion:

(a) $M + T$ (b) $\sqrt{(M^2 + T^2)}$

(c) $\frac{1}{2}(M + T)$ (d) $M^2 + T^2$

✔ **Answer: (b) $\sqrt{(M^2 + T^2)}$**

Explanation: $T_e = \sqrt{(M^2 + T^2)}$.

Q12. Stiffness of two springs in series:

(a) $k_1 + k_2$ (b) $k_1k_2/(k_1 + k_2)$

(c) $k_1 - k_2$ (d) $\sqrt{(k_1k_2)}$

✔ **Answer: (b) $k_1k_2/(k_1 + k_2)$**

Explanation: $1/k = 1/k_1 + 1/k_2$.

Q13. Stress in the wire of a close-coiled spring:

(a) Bending (b) Torsional shear

(c) Tensile (d) Compressive

✔ **Answer: (b) Torsional shear**

Explanation: The wire is in torsion.

Q14. M = 3, T = 4 kN·m — equivalent BM:

(a) 3.5
(b) 4
(c) 5
(d) 7

✔ **Answer: (b) 4**

Explanation: ½(3 + 5) = 4.

Q15. Spring deflection varies with wire diameter as:

(a) d
(b) 1/d
(c) $1/d^3$
(d) $1/d^4$

✔ **Answer: (d) $1/d^4$**

Explanation: $\delta \propto 1/d^4$.

SSC JE / UPSSSC JE Previous Year Questions — Torsion of Circular Shafts

Q1. Polar moment of inertia of a hollow shaft:

SSC JE (Previous Year)

(a) $\pi(D^4 - d^4)/32$
(b) $\pi(D^4 - d^4)/64$
(c) $\pi(D^3 - d^3)/16$
(d) $\pi(D^2 - d^2)/4$

✔ **Answer: (a) $\pi(D^4 - d^4)/32$**

Explanation: $J = \pi(D^4 - d^4)/32$.

Q2. Maximum shear stress in a shaft occurs at:

SSC JE (Previous Year)

(a) Centre
(b) Outer surface
(c) Half radius
(d) Everywhere equal

✔ **Answer: (b) Outer surface**

Explanation: $\tau = Tr/J$, maximum at r = R.

Q3. Polar section modulus of a solid shaft:

SSC JE (Previous Year)

(a) $\pi d^3/16$
(b) $\pi d^3/32$
(c) $\pi d^4/32$
(d) $\pi d^2/4$

✔ **Answer: (a) $\pi d^3/16$**

Explanation: $J/R = \pi d^3/16$.

Q4. For the same weight, compared to a solid shaft, a hollow shaft is:

SSC JE (Previous Year)

(a) Weaker
(b) Stronger
(c) Equal
(d) Less stiff

✔ **Answer: (b) Stronger**

Explanation: Material at outer region.

Q5. In T/J = τ/R, J is:

SSC JE (Previous Year)

(a) Moment of inertia

(b) Polar moment of inertia

(c) Section modulus

(d) Joule

✔ **Answer: (b) Polar moment of inertia**

Explanation: Polar moment of inertia.

UPSSSC ALERT

Chapter 6: Thin-walled Pressure Vessels

Thin-walled Pressure Vessels – Basic Concept

A cylindrical or spherical vessel whose wall thickness is less than 1/20 of its diameter (t < d/20) is a **thin vessel**. Boiler drums, gas cylinders, pipes and tanks are examples. Internal pressure produces hoop and longitudinal stresses in the wall.

Understand in Detail — Thin-walled Pressure Vessels

6.1 Thin pressure vessels — gas cylinders and boilers

An LPG cylinder, a boiler drum, a water pipeline, a balloon — all have internal pressure p pushing the wall outward in every direction. If the wall thickness (t) is very small compared to the diameter (d) (**t < d/20**), the vessel is considered **thin** and the stress can be assumed **uniform** through the wall thickness.

A cylindrical vessel can burst in two ways: (1) **along its length** (on a longitudinal line) — the stress resisting this acts around the circumference = **circumferential or hoop stress (σ_c)**; (2) **around its circumference** (on a transverse line, into two pieces) — the stress resisting this acts along the length = **longitudinal stress (σ_L)**.

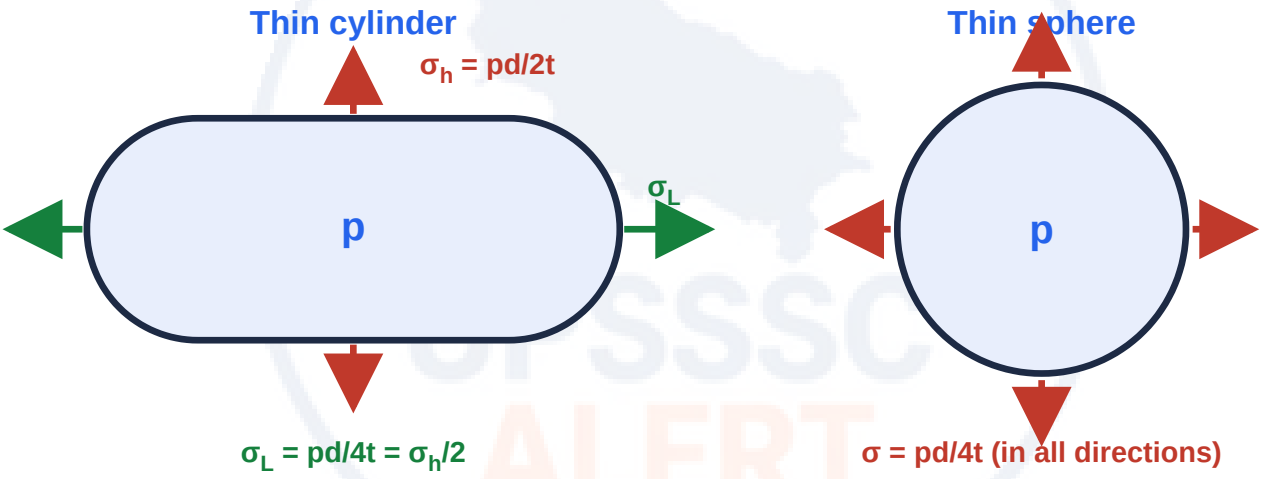


Fig 3.6: Stresses in a thin cylinder and a thin sphere

Derivation: Hoop and longitudinal stresses

1. **Hoop:** cut the cylinder lengthwise in half. Bursting force = p × d × L (projected area) ; resistance = $\sigma_c \times 2tL$ (two walls)
2. $\boldsymbol{\sigma_c = pd/(2t)}$
3. **Longitudinal:** cut transversely. Force = p × $(\pi/4)d^2$; resistance = $\sigma_L \times \pi\, d\, t$
4. $\boldsymbol{\sigma_L = pd/(4t) \rightarrow \sigma_c = 2\sigma_L}$ — hoop stress is double!
5. **Spherical vessel:** same in every direction → σ = pd/(4t) — half the hoop stress of a cylinder of the same diameter
6. **Maximum shear** (cylinder, in-plane) = $(\sigma_c - \sigma_L)/2 = pd/(8t)$

Consequences: (1) hoop stress is larger, so a cylinder bursts **along its length (on a longitudinal line)** — that is why a water pipe split by freezing in winter splits lengthwise. (2) In a boiler the **longitudinal joint** (which carries the hoop stress) is made stronger (higher efficiency η_L): $\sigma_c = pd/(2t\eta_L)$, $\sigma_L = pd/(4t\eta_c)$. (3) A **spherical vessel** manages with half the thickness at the same pressure — that is why large gas holders are spherical.

Change in volume (cylinder): hoop strain $e_c = (\sigma_c - \mu\sigma_L)/E = (pd/4tE)(2 - \mu)$; longitudinal $e_L = (pd/4tE)(1 - 2\mu)$; **volumetric strain $\boldsymbol{e_v = 2e_c + e_L = (pd/4tE)(5 - 4\mu)}$**. For a sphere $e_v = 3e = (3pd/4tE)(1 - \mu)$.

Example: A boiler of 1.5 m diameter works at 1.2 N/mm². If allowable tensile stress is 100 N/mm², minimum thickness? (Assume 100% joint efficiency)

Solution:

1. Hoop stress governs: $t = pd/(2\sigma) = 1.2 \times 1500/(2 \times 100) =$ **9 mm**
2. Check: $\sigma_L = 1.2 \times 1500/(4 \times 9) = 50$ N/mm² < 100 ✔
3. If spherical: $t = pd/(4\sigma) =$ **4.5 mm** — half

1. Key Points

- **Hoop (circumferential) stress:** $\sigma_h = pd/2t$; it acts on the longitudinal section and tends to split the cylinder along its length.
- **Longitudinal stress:** $\sigma_L = pd/4t = \sigma_h/2$; it tends to separate the cylinder into two parts around the circumference.
- **Thin sphere:** σ = pd/4t in all directions; so for the same pressure and diameter a sphere needs half the thickness of a cylinder.
- **Maximum shear stress (cylinder):** $\tau_{max} = (\sigma_h - \sigma_L)/2 = pd/8t$ (in-plane).
- **Design:** thickness from hoop stress: $t = pd/(2\sigma_t\eta_L)$, where η_L is the efficiency of the longitudinal joint. Longitudinal stress uses the circumferential joint efficiency η_c.
- **Volumetric strain (cylinder):** $\delta V/V = (pd/4tE)(5 - 4\mu)$; for a sphere $\delta V/V = (3pd/4tE)(1 - \mu)$.
- **Radial stress:** it is neglected in thin vessels; in thick cylinders it is found by Lamé's equations.

- **Thick cylinder:** $t > d/20$; hoop stress is maximum at the inner surface and radial stress varies from $-p$ inside to zero outside (Lamé).
- **Change in diameter and length:** $\delta d/d = (pd/4tE)(2 - \mu)$; $\delta L/L = (pd/4tE)(1 - 2\mu)$.
- **Wire-wound cylinder:** winding wire under initial tension produces compressive hoop stress, increasing pressure capacity (gun barrels).

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Hoop stress	**$\sigma_h = pd / 2t$**
Longitudinal stress	**$\sigma_L = pd / 4t$**
Ratio	**$\sigma_h : \sigma_L = 2 : 1$**
Thin sphere	**$\sigma = pd / 4t$**
Max in-plane shear	**$\tau = pd / 8t$**
With joint efficiency	**$\sigma_h = pd/(2t\eta_L)$, $\sigma_L = pd/(4t\eta_c)$**
Volumetric strain of cylinder	**$(pd/4tE)(5 - 4\mu)$**
Thin vessel condition	**$t < d/20$**

3. Detailed Explanation

- **Derivation of hoop stress:** bursting force on half cylinder = $p \times d \times L$, resisted by two walls: $2 \times \sigma_h \times t \times L$. Equating gives $\sigma_h = pd/2t$.
- **Derivation of longitudinal stress:** end force = $p \times \pi d^2/4$, resisting area $\approx \pi d \times t$; so $\sigma_L = pd/4t$.
- **Why a cylinder splits lengthwise:** the hoop stress is double, so the crack runs longitudinally; hence the longitudinal seam is made stronger.
- **Hemispherical ends of gas cylinders:** spherical ends have lower stress, so a thinner wall suffices.
- **Volumetric strain of a sphere:** $\delta V/V = 3 \times$ diametral strain $= (3pd/4tE)(1 - \mu)$.
- **Codes:** in India boilers follow IBR and vessels IS 2825; a corrosion allowance is added to the thickness.

4. Comparison Table — Thin Cylinder vs Thin Sphere

Basis	Cylinder	Sphere
Maximum stress	pd/2t (hoop)	pd/4t
Stress directions	Hoop ≠ longitudinal	Equal in all directions
Thickness needed (same p, d)	t	t/2
Examples	Boiler drum, pipes	Gas storage tanks

5. Solved Numerical Problems

Example 1. A cylinder of 1 m diameter and 10 mm thickness has internal pressure 2 N/mm². Find the hoop and longitudinal stresses.

Solution:

1. $\sigma_h = pd/2t = 2 \times 1000/20 =$ **100 N/mm²**
2. $\sigma_L = pd/4t =$ **50 N/mm²**

Example 2. A 1.5 m boiler works at 1.2 N/mm²; allowable stress 90 N/mm², longitudinal joint efficiency 80%. Find the thickness.

Solution:

1. $t = pd/(2\sigma_t\eta) = 1.2 \times 1500/(2 \times 90 \times 0.8)$
2. $t = 1800/144 =$ **12.5 mm**

Example 3. A 2 m thin sphere withstands 1 N/mm²; allowable stress 100 N/mm². Find the thickness.

Solution:

1. $t = pd/4\sigma = 1 \times 2000/400 =$ **5 mm**

Example 4. A cylinder (d = 800 mm, t = 8 mm) has p = 1.5 N/mm²; E = 200 GPa, μ = 0.3. Find volumetric strain.

Solution:

1. $pd/4tE = 1.5 \times 800/(4 \times 8 \times 200000) = 1.875 \times 10^{-4}$
2. $\delta V/V = 1.875 \times 10^{-4} \times (5 - 1.2) =$ **7.125×10^{-4}**

Example 5. A cylinder of 1.2 m diameter and 12 mm thickness has allowable hoop stress 100 N/mm². Find the maximum pressure.

Solution:

1. $p = 2t\sigma/d = 2 \times 12 \times 100/1200 =$ **2 N/mm²**

6. Frequently Asked Facts

- $\sigma_h = pd/2t$, $\sigma_L = pd/4t$; ratio 2 : 1.
- Thin sphere $\sigma = pd/4t$; half the cylinder's thickness is enough.
- Thin vessel: $t < d/20$.

7. Note (Confusing Facts)

- Hoop stress acts on the longitudinal section, so the cylinder bursts along its length.
- In thickness design, longitudinal joint efficiency is used with hoop stress.
- Radial stress in thin vessels is negligible and taken as zero.

Remember: One line: hoop = 2 × longitudinal. Assertion–Reason: (A) A boiler's longitudinal seam is a butt joint. (R) Hoop stress is greater. — Both true; R explains A.

30 Most Important One-Liner Facts — Thin-walled Pressure Vessels

1. Condition for thin vessel?

— $t < d/20$.

2. Hoop stress?

— $pd/2t$.

3. Longitudinal stress?

— $pd/4t$.

4. Ratio hoop : longitudinal?

— 2 : 1.

5. Stress in thin sphere?

— $pd/4t$.

6. Cylinder bursts along?

— Its length.

7. Max in-plane shear?

— $pd/8t$.

8. Radial stress in thin vessel?

— Negligible.

9. Equations for thick cylinder?

— Lamé's.

10. Sphere thickness for same p, d?

— Half of cylinder.

11. Another name for hoop stress?

— Circumferential stress.

12. Hoop stress acts on?

— Longitudinal section.

13. Thickness design formula?

— $t = pd/(2\sigma\eta_L)$.

14. Efficiency used for longitudinal stress?

— Circumferential joint.

15. Volumetric strain of cylinder?

— $(pd/4tE)(5 - 4\mu)$.

16. Volumetric strain of sphere?

— $(3pd/4tE)(1 - \mu)$.

17. Examples of thin vessels?

— Boiler, gas cylinder, pipes.

18. Hoop stress tends to?

— Split cylinder lengthwise.

19. Longitudinal stress tends to?

— Separate cylinder circumferentially.

20. Why spherical storage tanks?

— Lower stress, less material.

21. Thick cylinder condition?

— $t > d/20$.

22. Max hoop stress in thick cylinder?

— Inner surface.

23. Diametral strain?

— $(pd/4tE)(2 - \mu)$.

24. Longitudinal strain?

— $(pd/4tE)(1 - 2\mu)$.

25. Benefit of wire winding?

— Higher pressure capacity.

26. Boiler design code?

— IBR.

27. Extra allowance in thickness?

— Corrosion allowance.

28. Max pressure from σ?

— $p = 2t\sigma/d$.

29. Diametral strain of sphere?

— $(pd/4tE)(1 - \mu)$.

30. Principal stresses in thin vessel?

— Hoop and longitudinal.

Practice MCQs — Thin-walled Pressure Vessels

Q1. Hoop stress in a thin cylinder:

(a) pd/t
(b) pd/2t
(c) pd/4t
(d) pd/8t

✔ **Answer: (b) pd/2t**

Explanation: σ_h = pd/2t.

Q2. Longitudinal stress in a thin cylinder:

(a) pd/2t
(b) pd/4t
(c) pd/t
(d) 2pd/t

✔ **Answer: (b) pd/4t**

Explanation: σ_L = pd/4t.

Q3. Ratio of hoop to longitudinal stress:

(a) 1 : 1
(b) 1 : 2
(c) 2 : 1
(d) 4 : 1

✔ **Answer: (c) 2 : 1**

Explanation: pd/2t ÷ pd/4t = 2.

Q4. Stress in a thin sphere:

(a) pd/2t
(b) pd/4t
(c) pd/8t
(d) pd/t

✔ **Answer: (b) pd/4t**

Explanation: pd/4t in all directions.

Q5. A vessel is considered thin when:

(a) t < d/20
(b) t > d/20
(c) t = d/10
(d) t < d/4

✔ **Answer: (a) t < d/20**

Explanation: t/d < 1/20.

Q6. A thin cylinder under excess pressure bursts:

(a) Circumferentially
(b) Along its length
(c) At the ends
(d) Anywhere

✔ **Answer: (b) Along its length**

Explanation: Hoop stress is higher.

Q7. Max in-plane shear stress in a thin cylinder:

(a) pd/4t
(b) pd/8t
(c) pd/2t
(d) pd/16t

✔ **Answer: (b) pd/8t**

Explanation: (pd/2t − pd/4t)/2.

Q8. For the same p and d, thickness of a thin sphere is ____ that of a cylinder:

(a) Double
(b) Half
(c) Equal
(d) One-fourth

✔ **Answer: (b) Half**

Explanation: Sphere stress is pd/4t.

Q9. d = 1 m, t = 10 mm, p = 2 N/mm² — hoop stress:

(a) 50 MPa
(b) 100 MPa
(c) 200 MPa
(d) 25 MPa

✔ **Answer: (b) 100 MPa**

Explanation: 2 × 1000/20 = 100.

Q10. Thick cylinders are analysed by:

(a) Lamé's equations
(b) Euler's formula
(c) Unwin's formula
(d) Bernoulli's equation

✔ **Answer: (a) Lamé's equations**

Explanation: Lamé.

Q11. Maximum hoop stress in a thick cylinder occurs at:

(a) Outer surface
(b) Inner surface
(c) Mid-thickness
(d) Uniform

✔ **Answer: (b) Inner surface**

Explanation: Lamé.

Q12. Diametral strain of a thin cylinder:

(a) (pd/4tE)(2 − μ)
(b) (pd/4tE)(1 − 2μ)
(c) (pd/4tE)(5 − 4μ)
(d) pd/2tE

✔ **Answer: (a) (pd/4tE)(2 − μ)**

Explanation: $\sigma_h/E - \mu\sigma_L/E$.

Q13. Winding wire on a cylinder produces:

(a) Compressive hoop stress
(b) Tensile hoop stress
(c) Shear
(d) Nothing

✔ **Answer: (a) Compressive hoop stress**

Explanation: Initial compression.

Q14. d = 1.2 m, t = 12 mm, σ = 100 MPa — maximum pressure:

(a) 1 MPa
(b) 2 MPa
(c) 4 MPa
(d) 0.5 MPa

✔ **Answer: (b) 2 MPa**

Explanation: 2tσ/d = 2.

Q15. Boilers in India are designed as per:

(a) IBR
(b) ASME
(c) BIS 456
(d) ISO 9001

✔ **Answer: (a) IBR**

Explanation: Indian Boiler Regulations.

SSC JE / UPSSSC JE Previous Year Questions — Thin-walled Pressure Vessels

Q1. In a thin cylinder, hoop stress is ____ the longitudinal stress: SSC JE (Previous Year)

(a) Half
(b) Equal
(c) Twice
(d) Four times

✔ **Answer: (c) Twice**

Explanation: 2 : 1.

Q2. Stress in a thin spherical shell: SSC JE (Previous Year)

(a) pd/2t
(b) pd/4t
(c) pd/t
(d) 2pd/t

✔ **Answer: (b) pd/4t**

Explanation: pd/4t.

Q3. Thickness of a boiler shell is decided by: SSC JE (Previous Year)

(a) Longitudinal stress
(b) Hoop stress
(c) Radial stress
(d) Shear stress

✔ **Answer: (b) Hoop stress**

Explanation: Hoop stress is maximum.

Q4. Longitudinal stress in a thin cylinder: SSC JE (Previous Year)

(a) pd/2t
(b) pd/4t
(c) 2pd/t
(d) pd/t

✔ **Answer: (b) pd/4t**

Explanation: pd/4t.

Q5. Volumetric strain of a thin cylinder:

SSC JE (Previous Year)

(a) (pd/4tE)(5 − 4μ)

(b) (pd/4tE)(5 − 2μ)

(c) (pd/2tE)(1 − μ)

(d) (3pd/4tE)(1 − μ)

✔ **Answer: (a) (pd/4tE)(5 − 4μ)**

Explanation: 2 × diametral + longitudinal strain.

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